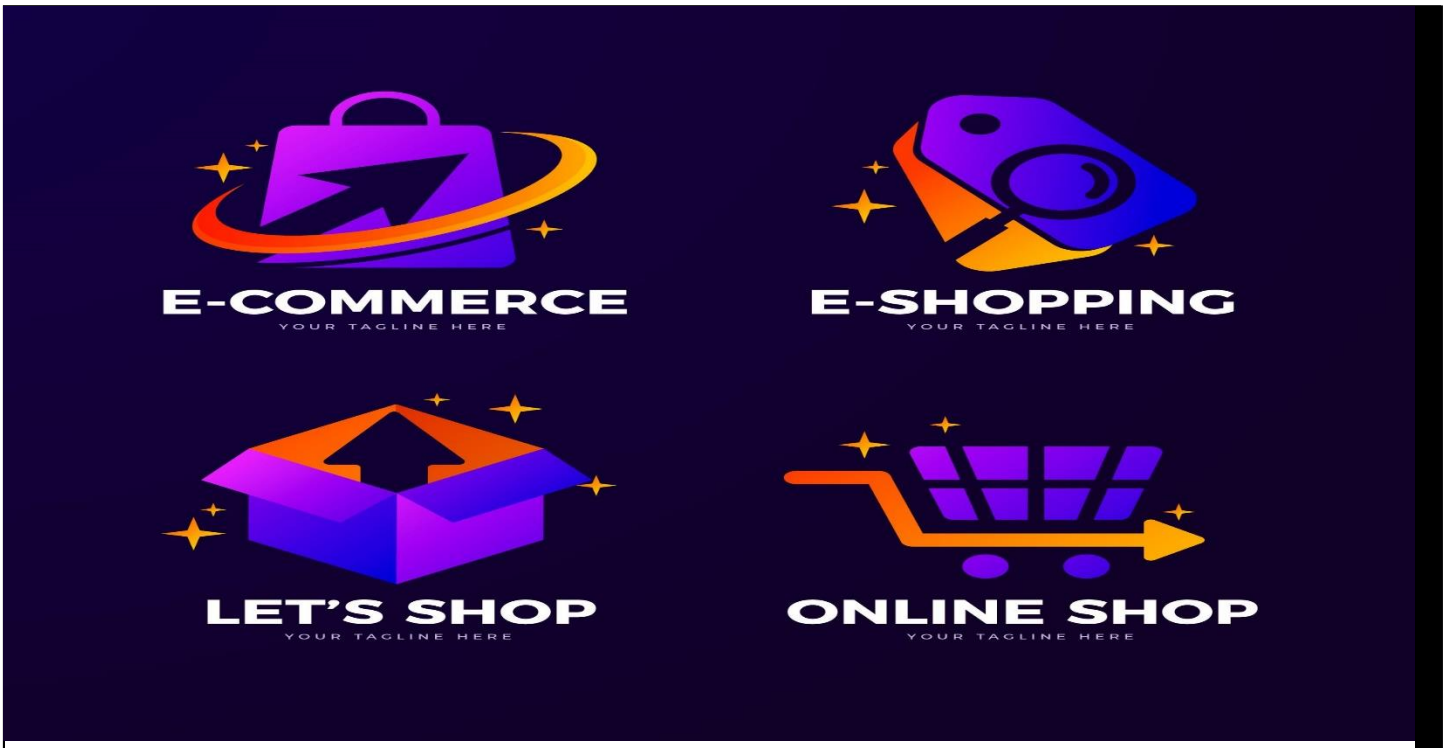
سُورَةُ الفَاتِحَةِ

بِسْمِ ٱللَّهِ ٱلرَّحْمَٰنِ ٱلرَّحِيمِ ١
ٱلْحَمْدُ لِلَّهِ رَبِّ ٱلْعَٰلَمِينَ ٢
ٱلرَّحْمَٰنِ ٱلرَّحِيمِ ٣ مَٰلِكِ يَوْمِ ٱلدِّينِ ٤
إِيَّاكَ نَعْبُدُ وَإِيَّاكَ نَسْتَعِينُ ٥ ٱهْدِنَا
ٱلصِّرَٰطَ ٱلْمُسْتَقِيمَ ٦ صِرَٰطَ ٱلَّذِينَ أَنْعَمْتَ
عَلَيْهِمْ غَيْرِ ٱلْمَغْضُوبِ عَلَيْهِمْ
وَلَا ٱلضَّآلِّينَ ٧



E-Commerce APP

Flutter - Web App (React-js)

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E-Commerce APP

Flutter

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ABBREVIATIONS AND TERMS

TAMK
- Tampere University of Applied Sciences

OS
- Operating system

iOS
- iPhone operating system

API
- Application Programming Interface

SDK
- Software development kit

Firebase
- Database management service from Google. Here you can store all your online data.

C and C++
- Programming languages used for software development

JavaScript
- A web-based programming language

Json
- JavaScript Object Notation

Array
- An object that contains multiple values.

Array index
- the order of a data in an array

Init function
- Init function runs for the initialization of the software

Stateful widget
- A stateful widget describes the part of user interface

Stateless widget
- the state which cannot be changed while the program is running

Query
- A method for requesting a data from database

1 INTRODUCTION

In this thesis, the Flutter technology is utilized to develop an e-commerce mobile application. E-commerce, or electronic commerce, is a technique of purchasing goods and services through the internet that is growing in popularity and effectiveness for online businesses. Ecommerce offers a number of benefits, the most significant of which is that it is inexpensive and convenient.

As a consequence, we may describe e-commerce as the process of buying and selling goods and services through the internet, whether inside a country or across the globe. These companies have their own mobile apps or websites where customers may explore a range of products and services.

They have a choice of payment alternatives if they purchase anything and need to pay for it, including (credit card, debit card, PayPal). utilizing a website or a smartphone app the order is shipped to the customer's address after confirmation. If required, customers can follow the status of their orders. All of these things happen on the internet. (A Review of Narrative Literature of e commerce, 2017). This application's main purpose and goal is to empower and assist a person in accessing e-commerce platforms so that they may do their own online shopping

2 BACKGROUND AND PURPOSE

2.1 Project Background

Electronic commerce, generally known as online business and trade, alludes to the acquiring and selling of products and services well over internet. In the 1960s, the term "ecommerce" was coined. With the growing usage of mobile devices, social media has emerged as an effective reinforcement of the web page's strength. (salesforce)

Ecommerce (electronic commerce) is the term used to describe all online activity along with the purchase and sale of goods and services. To put it differently, ecommerce is the process of doing transactions over the internet. Ecommerce has existed since 1994, when Phil Branden Berger bought the first product online, Sting's "Ten Summoner's Tales." By 2021, an estimated 27.2 percent of the world's population will be shopping online. (Oberlo, no date)

To deliver information to individuals, several innovative strategies and innovations in e-commerce applications have been established. Globally, online shopping has experienced substantial growth. Online shopping is used by 21.8 percent of the world's population. You're missing out on a huge group of potential purchasers if you're not selling online. 73% of customers browse across several channels, including Google, social media, and email. (salesforce)

2.2 Project Objectives

In recent years, online purchasing has become increasingly popular. People all around the world prefer to buy things online rather than in stores, which is why Amazon and other similar online retailers are not only successful but also increasing interest in online commerce. (Darrell, 2011)

In today's world, though, innovation is assisting us in breaking down some of these barriers by using computer frameworks to aid us with various forms of labour. As we all know, the world of online purchasing is quickly developing. As

a result, e-commerce apps have already developed various new tactics and technologies for giving information to individuals. These ideas and technologies, however, provide a number of challenges and concerns for people. (sgrlaw)

2.3 Comparative Analysis

Existing Solutions	**Proposed Solution**
Assists in the improvement of our standard, carefully crafted products for distribution within the showcase.	More attractive user interface.
The framework allows a buyer to quickly explore or search for an item.	Faster Server-to-client-side response.
It provides an easy way to purchase items directly from sellers without the use of a third-party interface	Tried to make the rural individuals fill out the description form which will provide more about their products.
	Tried providing a better and user friendly environment than the existing one.

3 ANDROID APPLICATION REQUIREMENTS AND DEVELOPMENT

3.1 Ecommerce Application Development

The history of e-commerce dates back more than 50 years, when early inventions such as EDI (Electronic Data Interchange) and teleshopping were shown in the 1970s, paving the way for today's e-commerce companies. The history of e commerce is inextricably linked to the history of the internet. When the internet was invented in the 1990s, online shopping became possible. Since then, Amazon has become one of the most popular e-commerce sites in the United States, with hundreds of e-commerce enterprises springing up as a result. (Everydayonlinemarketing, no date)

Michael Aldrich demonstrated e-shopping by connecting a phone connection to an adjustable TV and a transaction-processing computer. (Aldricharchive, no date)

While desktop spending has remained stable throughout the year, the worldwide market for online business transactions has grown at an annual rate of 33.7 percent since 2016. This happens as a result of customers' continuous preparation with each screen tap. The creation of mobile applications for e commerce enterprises appears to be a long-term trend, according to simple reasoning. Nothing suggests that the worldwide public will soon reject smartphones. Diverse internet shopping, on the other hand, is predicted to reach $3.5 trillion by 2021. (Shopify, no date)

The e-Commerce business is experiencing the most notable growth of adaptable arrangements being prompted by the need to meet client needs in the context of rising online sales. Customers want the ability to purchase items and services at any time and from any location, without having to travel. Customers choose various e-commerce applications or websites to purchase things because they advance faster, give more comfort, and are more flexible.

For As a mobile application developer, it's important to remember the distinctions between mobile development and more traditional software development. Consider how varied the settings for utilizing mobile apps vs desktop computers can be, for example, recognizing the constraints and complexities of a touch based user interface, and considering the unique security vulnerabilities that mobile technologies provide. A mobile developer should also be mindful of the importance of user experience and the fact that most users are acclimated to a certain operating system's design guidelines. (creativebloq, no date)

3.2 Android App development

Android app development is the process of creating mobile apps for devices using the Android operating system. Languages like C, object-oriented languages (like Java), and the SDK (Android Software Development Kit) are used to create Android applications. Android was initially introduced in 2009, and it is based on the Java programming language. In today's world, Android dominates the whole portable app sector, with the majority of the market share and a global clientele.

3.3 Functional Requirements

The functional requirements for this application that are necessary to complete this project are as follows:

• Application must start when client clicks on app icon

• Application must show the home page of the application first

• Application must provide the selected category’s content list for the client

• Client must be able to view the products

• Client must be able to view the products image

• Application must provide the selected product’s description for the client

• Client must be able to add the product to the cart

• Application should add the product to the customer’s shopping cart.

• Application must provide the customer’s information form for the client.

• Client must send the provided credentials to the application.

3.4 Non-Functional Requirements

The non-functional requirements for this application that are necessary to complete this project are as follows:

• Application should have an interface to display products to the customer.

• Application should have an interface to display list of categories

• Application should be able to display list of categories, so customers easily click on them

• Application should be able to display a product's description

• Application should have an interface for the customer to add a product to the shopping cart

3.5 GUI (Graphical User Interface)

Main App splash screen and user login screen

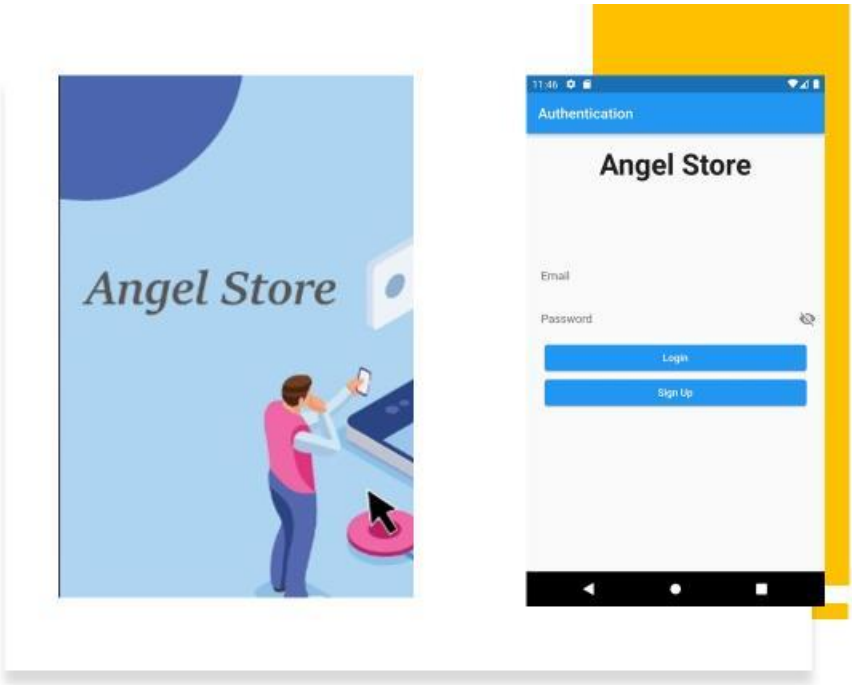


Figure 1: Customer login and splash screen

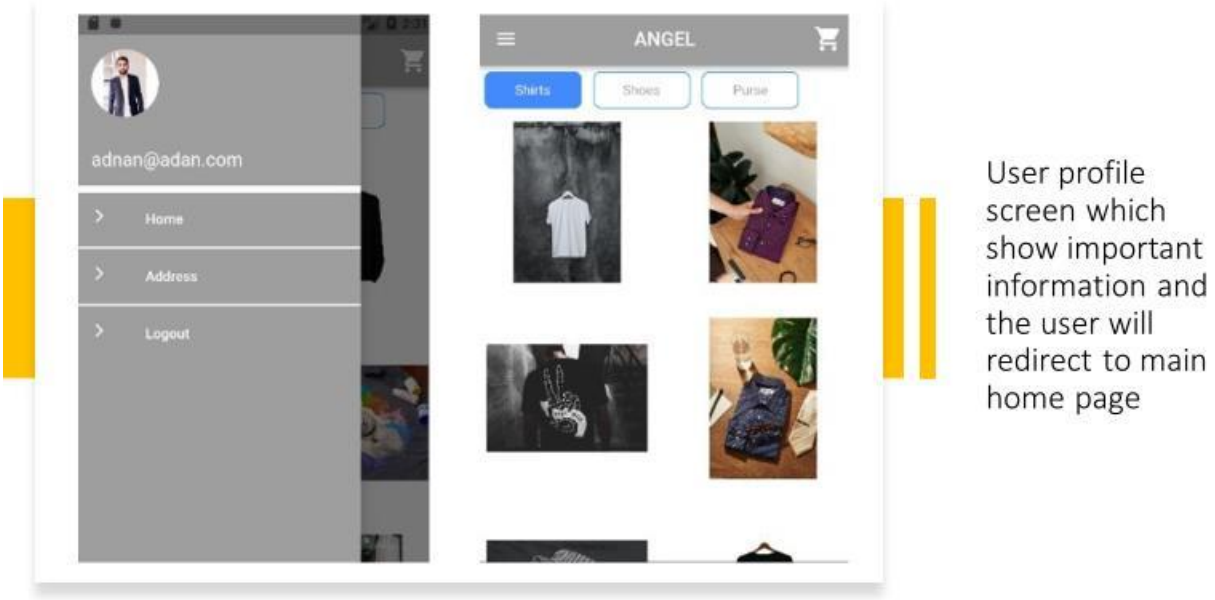


Figure 2: Profile and main screen

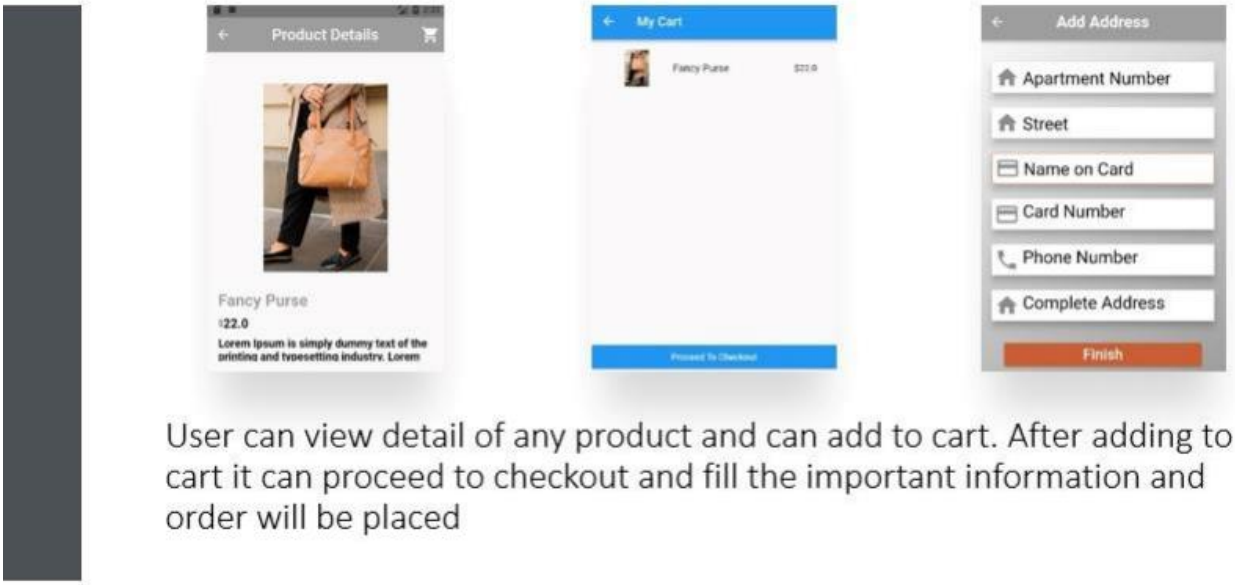


Figure 3: Order processing

4 TECHNOLOGY

4.1 Flutter

This application is built on Flutter technology. Flutter offers several advantages because it is a Google product. The following is a quick overview of flutter.

The app development process is revolutionized by Flutter. You can design, test, and publish stunning mobile, web, desktop, and embedded apps by writing one time code. Google built Flutter as an open-source project. It's used to make hybrid apps for Android, iOS, Linux, macOS, Windows, Google Fuchsia, and the web from a single codebase. (Flutter, no date)

You can develop and iterate quickly using Hot Reload. You'll observe changes very immediately after updating the code, with no loss of state. Maintain every pixel to create unique, responsive designs that look and feel great on every device.

Every component in flutter is referred to as a widget. It is entirely widget based. Stateless or stateful widgets are available. There are several states in a stateful widget, not just one. They have the ability to modify their state in response to user interest. A Stateless widget, on the other hand, is a static widget that is utilized for static data or functionality. It is devoid of any state. We need to utilize it in some circumstances because we work with static data. (AppDividend, no date)

Flutter is an open-source project, anybody may use and contribute to it at any level. It's better to use it and offer your passion for Google technology by becoming a Google contributor. Flutter has a massive user base since it is a Google product, and Google is the most widely used search engine on the planet.

Flutter outperforms other hybrid solutions in terms of cost. It's one of the many advantages of flutter. Flutter developers may encrypt app data with both asymmetric and symmetric keys using the iOS SecKey API and the Common Crypto library. Flutter code is written in Dart, which has many cryptos and

encrypts libraries that employ various cryptographic hashing and encryption techniques.

4.2 Android

Android is a multitasking mobile operating system that runs on smartphones, tablets, readers, televisions, and even domestic robots. Google purchased and marketed this operating system, which was developed by 'Android Inc' and based on Linux. Android's introduction as an operating system (OS) in 2008 was a spark, and it quickly became popular among smart gadgets. Modern smartphones and tablets might be considered stash minicomputers thanks to this operating system.

By 2020, the Android platform will have about 35 billion users worldwide, making it the most popular mobile operating system on the planet. A large number of smart gadgets run on this operating system (like phones, tablets, and smartwatches). Every company organization or brand in the world nowadays need an e-commerce Android app in order to grow their business, money, and popularity. (Android, no date)

Android is a very adaptive and engaging system, and a basic acquaintance takes less than an hour. Because there are so many essential programs accessible, any customer may easily configure OS settings. You can modify everything past recognition: if you don't like the look, symbols, or ringtone, simply go to the Google Play Store, download a significant program, and quickly customize everything to your desire

4.3 Dart Language

Dart is a client-oriented programming language that enables rapid app development across all platforms. Its primary goal is to create one of the most productive languages on a variety of platforms. Both the server and the user will benefit from it. The Dart SDK includes the Dart VM compiler as well as the dart2js tool, which generates the JavaScript version of a Dart Script so that it may be

executed on sites that don't support Dart. It is a very similar object-oriented programming language to C++. Dart is a popular programming language for creating single-page websites and online apps. (Dart, no date).

Dart is designed to provide logic and, as a result, a beautiful user interface. For mobile, desktop, and backend apps, compile specialized machine code. Alternatively, for web use, compile to JavaScript.

Benefits

The first advantage is that it is a very easy language to pick up. If any of us are familiar with C or C++, we can quickly learn this language because the grammar is comparable. The vast community, which provides extensive documentation, is the second item to note. And because it's a Google product, they make it as simple as possible. The third advantage is its excellent performance. Dart-based programs are faster to execute than JavaScript-based programs. (Code Carbon, 2020)

Drawback

Of course, we must consider the drawbacks in addition to the rewards. The first disadvantage is that it has limited resources, making it harder to find solutions to problems. This is due to a lack of larger and more cohesive development communities to assist you.

Dart is currently in development. That's fantastic, and there's nothing "wrong" with Dart, but if you start writing in Dart today, the API may change, or things may not be fully or accurately documented, and the quantity of knowledge accessible on the web may be less than, example, the amount of information available on jQuery. (Code Carbon, 2020)

4.4 Google Firebase

Firebase is a Google program that allows developers to create apps for a variety of platforms, including iOS, Android, and web-based apps. Firebase is a quick tool for developing things that would take a long time in a traditional database

system, such as bespoke APIs. Firebase has built-in APIs that allow us to construct applications quickly. One of the nicest features of Firebase is that it gives us tools for tracking business reports and experimenting with different goods.

The Firebase database is a real-time database that allows clients to immediately access data. Even if you are not connected to the internet, you can still use your program with all of your prior data sets. Firebase allows users to use data in a secure and secured manner.

The Firebase database is a NOSQL database, similar to the MYSQL database used in PHP applications. The real-time database API was created exclusively for rapid operations and capabilities. (Firebase, no date).

4.5 Android Studio

It's a Google-created integrated development environment (IDE). The goal of Android is to speed up progress and make it easier for users to create high-quality apps for Android devices. Commonly used operating frameworks, such as Mac and Windows, allow variations of this IDE. It also provides a development kit and plugins for developers. cross-platform application support (IDE)

5 TECHNICAL IMPLEMENTATION

5.1 Planning

The initial stage in our project's technological execution is to plan what we want to build that will benefit our society. We've attempted to question everything that may be touched by the program up to this point:

• Will the development take place in stages or all at once?

• How much time will it take?

• Will there be new security challenges?

Finally, we want to create an ecommerce software that allows users to purchase items without having to go to an actual store or shop. This is an excellent concept for easing the difficulties faced by many folks who are unable to go to the market to purchase goods. An entire store is there at their fingertips; all they have to do is scroll and choose the things that best suit their needs.

5.2 MVVM (Model-View-View Model)

The app's current design is built on the MVVM paradigm, which employs a well known provider pattern. Views in this design closely monitor the view Model and react to any changes that are detected. We have a home-view that is watching Data View Model, and the Data View Model has a reference to the model in our example.

In the Data View Model, we have a method for filling categories that is responsible for adding categories to the list of categories and informing the listener, much like populating categories. The Home-view has a Horizontal List View widget that monitors the category list and updates the content of the list View whenever the list is modified.

The same goes for categorizing items and filling them. The Data View Model has three methods: getShoes(), getPurse(), and getShirts(). When we choose a category, it queries the items in that category and alerts the Grid view widget,

which oversees displaying products. The Selected Product reference variable in the DataViewModel is responsible for displaying product details on the product details page.

When we pick a product from the Home Page, we update the reference variable for that product, which is visible in the product details page, and we alter the content of the detail page appropriately. In DataViewModel, we additionally keep track of the listOfCartProducts. We update the listOfCartProducts in DataViewModel when we add a product from the details page to the cart. This variable is tracked in the cart Screen and updates the listView widget when the list changes.

Furthermore, we manage users using Firebase authentication. Orders may only be placed by authenticated users. We built a project for Angel for firebase configuration, and under Project, we developed an app for Android and iOS based on the package name of Android and the bundle id of iOS, respectively. For Android and iOS, Firebase generates a google-services.json and info-plist file, accordingly. We placed the files in the appropriate directories and then implemented the authentication system using the Firebase APIs

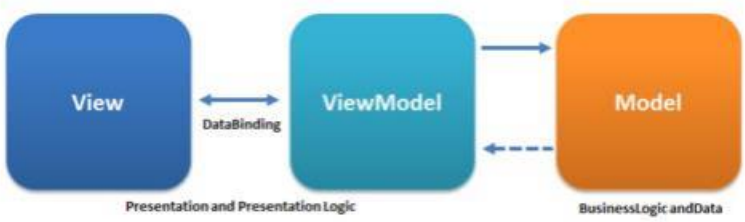


Figure 4: Model–view–viewmodel

5.3 Configuration and Customization

It's now time to install the programmer and begin configuring the functionalities that may be used right away. In fact, this step should always come first before developing any procedures or rushing to customize any module.

• The first step is to install and configure flutter. After that, we'll install it by following the instructions on the official flutter website.

• The second step is to install and configure Android Studio, after which we will install the flutter and dart plugins to use flutter in Android Studio.

• We built up an emulator in Android Studio to run and test our app in this phase.

• The fourth step is to create a home screen. Then we used dart language to create the home screen.

• We developed items and their product detail pages after designing the home screen. Similarly, we have created each design component separately.

• To pick numerous things at once, we have provided add to cart capability to our app. The add to cart feature also allows you to buy numerous items at once.

• We've introduced a login and registration form, and users must first register and login before placing an order. This was accomplished using firebase and flutter.

5.3.1 Flutter Firebase working

When a client (iOS or Android) requests a firebase API, the client sends its query to the firebase database, which then responds to the firebase API and the client (IOS or Android).

In Flutter, this is a lib folder where we can handle all our app's screens. This folder also contains the functioning section. This folder is responsible for 90% of our app development.

• **Pubspec.yaml** manages all our library files. This is a very private section of our software. It is also space constrained.

• Two responsible libraries for firebase integration are

1.

firebase_core: ^1.16.0

2.

firebase_auth: ^3.3.17

• Flutter asset folder, where we keep track of all the assets for our flutter app. Images, static data, videos, and other assets

• The IOS folder is responsible for all IOS development operations. Flutter is a hybrid technology, which means it can manage several platforms.

• It manages all the tasks required for Android development, same as IOS.

5.3.2 Firebase Configuration for Android and IOS:

To use Firebase with Flutter, we must first create a Firebase console project. We've established a project called Angel on Firebase. After that, we added packages for Android and iOS.. We inserted a google json file for Android and an info plish file for iOS in the root directory of both Android and iOS. This is how we set up our Firebase application for iOS and Android. We've also included packages in the pub spec files for authentication.

We received the project id, which is essentially our project's unique identification. We also issued a project number, which we may use to configure integrations with Firebase or other third-party services. We also gained the public facing name, which will appear on emails sent to users after they create an account in our application

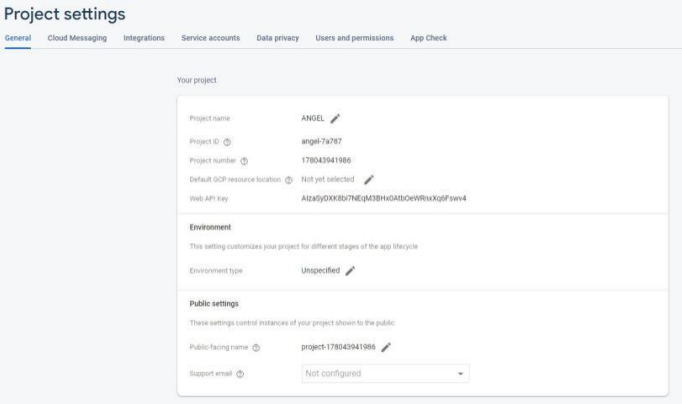


Figure 5: Firebase project settings

We ran into the following issues during configuration and customization.

• We placed the google services json file in a different location from the root directory while configuring the application for Android. As a result, when we test an application, we get unexpected problems. However, when we place the google services json file in the root directory of Android, it does not work.

• We had an authentication problem with the application while managing sessions. When a user logs out of an application, they will be sent to the home screen without having to log in again. This is a problem with authentication. We have resolved in our app.

• We have a problem with password letters. When we enter a password with fewer than six letters, it is not checked, and no error is displayed. We've corrected the problem, and now the password must contain more than six letters.

• When we add a product to a cart, it appears in the cart screen. This has been a huge problem for us. This problem was handled by examining indexes in an array. An array does not hold indexes. As a result, the cart displays an empty value.

• Users do not need to log in before placing an order. We were able to resolve this problem by utilizing session and authentication. Users must

now be logged in before placing an order or seeing the main home page.

6 CODE EXPLANATION

6.1 Assistance from the open sources and friend

I received design assistance from one of my mates as well as other sources. The Shipping details page, intro screens, and Splash screen were created by a friend, and the design was pulled from site on the internet and integrated into one.

We got a reasonably decent open-source code from GitHub for designing part of the code and used parts of the code from that open source and made adjustments to it by watching YouTube tutorials to add some other components as well. (MonsieurZbanowanYY, no date)

We used the cart component with the help of YouTube instead of the search bar on the top right side of the application. We've also created a shipping detail section where we ask for buyer information such as address, phone number, and so on. (The Flutter Way, 2020)

6.2 Whole Application background Function calling

When a user opens an app for the first time, we set a background function to call once when the app is opened by a new user. It has 3 to 4 guides to keep users entertained. When the home screen is opened, it calls the home screen widget, which then calls the state widget to receive product data and display it in a grid on the home screen.

Navigator is a tool that allows us to navigate data by category. When a user taps

on a product, we acquire its array index and retrieve the desired object detail. When a user taps on a product (which are taken from an open source), the product's details screen opens. On the detailed screen, we match a product's id with the array and display data when the id matches the array index. (Unsplash) On the detail screen, we have fields for description, price, add to cart, and buy now. It will be added to the cart when the user clicks on add to cart. When a user clicks the buy now button, they will be taken to the checkout page. If the user is logged in, he or she can place an order; if not, the user must first log in to their account, then register for an app, and then log in

```
void login() {
  AuthenticationHelper()
      .signIn(email: usernameController.text, password:
passwordController.text)
      .then((result) {
    if (result == null) {
      // Navigator.pop(context);
      Navigator.pushReplacement(context,
          MaterialPageRoute(builder: (context) => CardScreen()));

       ScaffoldMessenger.of(context).showSnackBar(SnackBar(
         content: Text(
           "Login Successfully",
           style: TextStyle(fontSize: 16),
         ),
       ));
    } else {
      ScaffoldMessenger.of(context).showSnackBar(SnackBar(
        content: Text(
          result,
          style: TextStyle(fontSize: 16),
        ),
      ));
    }
});
```

Figure 6: Login Code

When the user adds products to the cart, he or she will check the cart to see which items have been added. He/she will inspect the items and proceed to checkout from here. When he/she taps to check out again, the login session is checked, and if he/she is not logged in, they must first log in before placing an order.

We also display products underneath the details screen of each product, allowing the user to select products from the same category. User interface processes are

managed by libraries. We just display a Thank You screen after the order is placed. Because our program has numerous screens, we will discuss each screen's workflow as well as the packages that are included in order to explain the code.

6.3 Home Screen

As our Home screen has following categories Shirts, shoes, and purse. Every product is associated with a specific category. We have made a dart file name as Angel_Home_Screen.Dart. In this file our main home categories are managed. Then we make a file name as prepare_data. In this file we make a class Data_View_Model.

In this class we are managing our all products, categories as well as cart. We are managing all the stuff in a single file in order to reduce no of files and optimization of code. In this class we make a method for each functionality. In this class we make get purse function in order to get all products in a purse. We make get shoe ad get shirts as well in a same logic.

We added loops for every category where are the products in a category are managed. If a category has 10 numbers of products, then loop move 10 time in order to show all the products for a specific category.

As we use provider patterns for this application, so the main class extends change notifier. By using this provider pattern listen about the list that are prepared in this class. When any change occur in this class then it will notify to provider about change. Basically, this is the best thing in providers for managing data in application.

In our application we are using init function. Init function we are calling our Data_View_Model class. Everything is prepared before running. Let’s suppose you open an application. You can see only 1 category products by default. When we click on other category notifier note the change and send to the listener that is call in init() function. Then home screen data index is updated and across index

specific category products are call and display in a home screen.

6.4 Login Registration Screen

In login and registration page we are using stateful widget. In Stateful widget we call its state. In Its state we are using scaffold widget. Scaffold is a layout of our whole screen which gives us many things like App Bar, Body and many more.

We are using column widget to align in vertically direction. In column we have a children. In Children we are using text fields. One for email and other for password.

We have used signInWithEmailAndPassword and for the registration we have used createUserWithEmailAndPassword API's.

When a user enters email and password. A query is generated which checks weather registered on firebase database or not. If credential are match it return a true and we redirect it into the screen where they enter detail.

When a user login to their account, we save its email and give it to unique key. When they press logout, its key will be removed and when they try to login again we will give them a new key again. In this way we authenticate our user.

We are managing session as well when a login to their account session start and when user logout an application it will be over. We are managing session by using shared preferences in flutter. Shared preference allow us to manage user session with a specific interval of time. We have added a package in pub spec in order to use in flutter.

shared_preferences: ^2.0.13

These packages are used in application to perform login and registration.

Firebase core provide us an access to the flutter database. Firebase core also allow us to perform initialization actions. On the other hand, Firebase auth allow us to authenticate user when they perform registration or Login. By using these packages, we can use firebase APIs for login or registration.

firebase_core: ^1.16.0

firebase_auth: ^3.3.17

6.5 Add to Card Screen

In Add to cart screen we are using stateful widget. In Stateful widget we call its state. In Its state we are using scaffold widget. Scaffold is a layout of our whole screen which gives us many things like App Bar, Body and many more. We are using column widget to align in vertically direction. In column we have a children. In Children we are using list view. In list view widget all the selected items are arranged.

Session check also included in add to cart functionality. If a user login to their account, then its session started then they will purchase product by using add to cart functionality. It is very good practice to use session in purchasing points to authenticate user.

We are fetching a data in our list view which index marked as true. In every item in list view, we are making edit and delete button. Below list view we are using elevation button to proceed to checkout. Elevation button has a on tap functionality. We set checkout page path in on tap functionality

7 Discussion and Further Development

Online purchasing has grown increasingly popular in recent years. People all over

the world prefer to purchase online rather than go to the store, which is why Amazon and other similar online stores are not only profitable but also generating interest in online business. Many new strategies and technologies have been created in e-commerce apps to offer users with information. Online commerce is experiencing considerable development all around the world.

We've provided answers, but they'll need some time and resources. This App will provide users with modules to help them with their task. The player's major goal in building the app was to prioritize quality above quantity. As previously indicated, more modules and functionality will be introduced soon to make the program more solid and adaptable.

The primary goal and aim of this program are to enable and assist people in using an e-commerce-based platform, which is widely utilized throughout the world for online shopping and purchase of commodities and products nowadays. As a result, individuals may enjoy, conveniently access, and purchase all things through internet shopping and feel fully integrated into society. Our effort will not end here; we will continue to improve the software and provide new features for those who are unable to perceive the world from any angle (completely blind people). We'll aim to include a feature that makes this app totally accessible to blind people; they'll be able to control it with their voice and dress themselves according to their preferences. We'll be adding additional features as time goes on.

REFERENCES

Abilitynet. 2013. How to use TalkBack in Android Jelly Bean. Read on

25.05.2022.

https://mcmw.abilitynet.org.uk/talkback-android-jelly-bean-4-2-2

Android Accessibility Help. 2022. Get started on Android with TalkBack. Read on 11.05.2022.

https://support.google.com/accessibility/android/answer/6283677?hl=en

AppDividend. no date. Flutter Mobile Development Guide For Beginners. Read on 08.05.2022.

https://appdividend.com/category/flutter/

Code Carbon. 2020. Major Advantages And Disadvantages Of Dart Language. Read on 15.05.2022.

https://codecarbon.com/pros-cons-dart-language/

Collin, C. 2021. 5 Critical Steps for Your Software Implementation Plan. Read on 11.05.2022.

https://www.softwareadvice.com/resources/software-implementation-plan/

Dart. No date. Dart programming language | Dart. Read on 05.05.2022.

https://dart.dev/

Darrell, R. 2011. The Future of Shopping. Harvard Business Review. Read on 08.05.2022.

https://hbr.org/2011/12/the-future-of-shopping

Everyday. No date. Essential reads: Where marketing meets strategy - Everyday Online Marketing. Read on 05.05.2022.

https://everydayonlinemarketing.com/

Firebase. No date. Firebase documentation: Learn the fundamentals. Read on 01.05.2022.

https://firebase.google.com/docs

Flutter. no date. Flutter: Build apps for any screen. Read on 10.05.2022.

https://flutter.dev/

GeeksforGeeks. No date. A computer science portal for geeks. Read on 06.05.2022.

https://www.geeksforgeeks.org

Google. 2022. Everything you need to build on Android. Read on 24.05.2022.

https://developer.android.com/studio/features

James, H. 2022. What is Java? Definition, Meaning & Features of Java Platforms. Read on 12.05.2022.

https://www.guru99.com/java-platform.html

Lockwood, B. no date. 10 Technology Challenges. Read on 03.05.2022.

https://sgrlaw.com/ttl-articles/10-technology-challenges/

Michael Aldrich Archive. No date. Welcome to the Michael Aldrich Archive. Read on 09.05.2022.

https://www.aldricharchive.co.uk/

MonsieurZbanowanYY. 2022. E-Commerce-App-UI-Flutter. Read on 27.05.2022.

https://github.com/MonsieurZbanowanYY/E-Commerce-App-UI-Flutter

Nazmun Nessa Moon , Shaheena Sultana, Fernaz Narin Nur & Mohd Saifuzzaman. 2017. A Literature Review of the Trend of Electronic Commerce in Bangladesh Perspective. Read on 23.05.2022.

https://globaljournals.org/GJMBR_Volume17/3-A-Literature-Review-of-the Trend.pdf

Oberlo. 2022. FULFILL YOUR DROPSHIPPING DREAMS. Read on 21.05.2022.

https://www.oberlo.com/

Pravine, V. no date. A 5-STEP PLAN FOR SOFTWARE IMPLEMENTATION

SUCCESS. Read on 18.05.2022.

https://www.chetu.com/blogs/technical-perspectives/software-implementation plan.php

Richard F. Schmidt. 2013. Chapter 19 - Software Implementation. Read on 25.05.2022.

https://www.sciencedirect.com/science/article/pii/B9780124077683000197

Salesforce, Inc. 2022. CRM Software; Cloud Computing Solutions - Salesforce EMEA. Read on 05.05.2022.

https://www.salesforce.com/eu/

Santex. No Date. 7 Steps for Software Implementation Success. Read on 20.05.2022.

https://santexgroup.com/blog/7-steps-for-software-implementation-success/

Shopify. No Date. Shopify: Start and grow your e-commerce business. Read on 17.05.2022.

https://www.shopify.com

The Flutter way. 2020. Online Shop App - Flutter UI - Speed Code. Published on 03.07.2020. Referred on 23.05.2022.

https://www.youtube.com/watch?v=XBKzpTz65Io

Unsplash . no date. Beautiful Free Images & Pictures | Unsplash. Read on 27.05.2022.

https://unsplash.com/

Wikipedia contributors. 2022. Model–view–viewmodel. In Wikipedia, The Free Encyclopedia. Read on 05.05.2022.

https://en.wikipedia.org/w/index.php?title=Model%E2%80%93view%E2%80%93viewmodel&oldid=1088169504

E-Commerce APP

Web App (React-js)

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CHAPTER 1:

Introduction :

1.1 Client/Organization profile:

• Organization/Client Name: Vardhaman Om Paper Product

• Location: Gate no-1010, opp nisarg hotel, mangdewadi , katraj pune-411046, Maharastra, India

• Email: ompaperproduct@gmail.com

About Organization:

Om Paper Products is one of the leading manufacturers of Aluminium Foil Containers, Printed Paper Cup, Plastic Disposable Trays etc. We direct all our activities to cater the expectations of customers by providing them excellent quality products as per their gratifications. Moreover, we follow moral business policies and crystal pure transparency in all our transactions to keep healthy relations with the customers.

For our accomplishment story, we are grateful to our owner, “Aditya (Owner)”, whose continual backing and direction have been useful to us for attaining exponential development in the current market.

1.2 Need for System:

Organization currently doing business without any publicity on online platform.

The client need to develop website for their organization for selling products.

The requirement about this website is mainly selling their product online.

- Seurity – User’s data will be secured with username and password

1.3Scope & Feasibility of Work:

The System “Ecommerce website” will be a website where manufacturers who are willing to sell product/goods online and similarly customer who wants to purchase products online. Website will be available for 24/7. User will be able to run any kind of business using website. Customers can purchase product very easily.

Feasibility of Work :

Feasibility is the determination of whether or not a project is worth doing the process followed by making this determination is called a feasibility study. This of determine if a project can and should be taken. Once it has been determined that a project is feasible, the analyst can go ahead and prepare the project specification which finalizes project requirements. Generally, feasibility studies are undertaken within right time constraints and normally culminate in a written and oral feasibility report.

1. Technical feasibility:

The technical issue usually raised during the feasibility stage of the investigation includes the following:

• Does the necessary technology exist to do what is suggested

• Do the proposed equipment's have the technical capacity to hold the data required to use the new system?

• Will the proposed system provide adequate response to inquiries, regardless of the number or location of users?

• Can the system be upgraded if developed?

• Are there technical guarantees of accuracy, reliability, ease of access and data security?

2. Operational feasibility:

Proposed projects are beneficial only ifthey can be turned out into information systems, which will meet the organization's operating requirements. Operational feasibility aspects of the project are to be taken as an important part of the project implementation. Some of the important issues raised are to test the operational feasibility of a project includes the following:

• Is there sufficient support for the management from the users?

• Will the system be used and work properly if it is being developed and implemented?

• Will there be any resistance from the user that will undermine the possible application benefits?

This system is targeted to be in accordance with the above mentioned issues. Beforehand, the management issues and user requirements have been taken into consideration. So there is no question of resistance from the users that can undermine the possible application benefits. The well planned design would ensure the optimal utilization of the computer resources and would help in the improvement of performance status.

3. Economic feasibility:

A system that can be developed technically and that will be used if installed must still be a good investment for the organization. In the economic feasibility, the development cost in creating the system is evaluated against the ultimate benefit derived from the new systems. Financial benefits must equal or exceed the costs. The system is economically feasible. It does not require any additional hardware or software.

1.4 Operating Environment –

H/w & S/w Hardware

required:

• RAM : 2 GB

• CPU Speed : 2.0 GHZ

• Hard Disk : 250 GB

Software required:

➢ **User side Software:**

• Windows

• Linux

• Technology: JAVA, SPRINGBOOT

• IDE : Spring Tool Suite

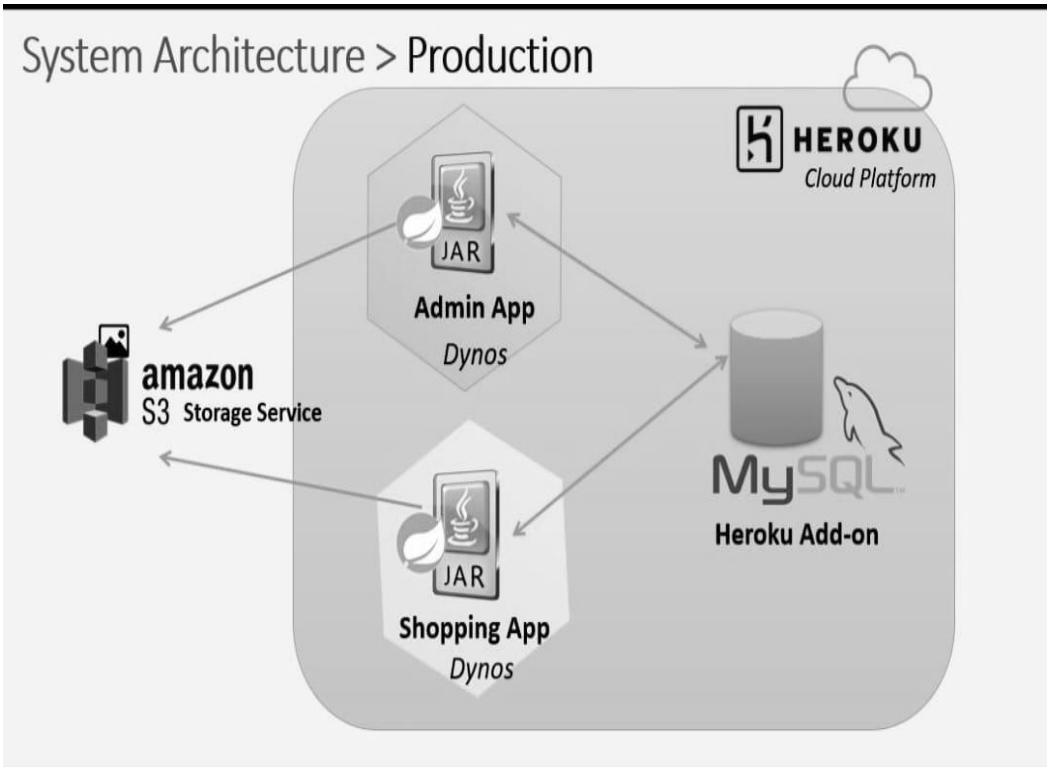
• Client Side Technologies: HTML, CSS, JavaScript , Bootstrap

• Server Side Technologies: Java

• Data Base Server: Sql

Operating System: Microsoft Windows/Linux

1.5Architectute Of System



1.6 Detailed Description of Technology Used:

Html5

Introduction: HTML stands for Hyper Text Markup Language. It is used to design web pages using a markup language. HTML is an abbreviation of Hypertext and Markup language. Hypertext defines the link between the web pages. The markup language is used to define the text document within the tag which defines the structure of web pages. HTML 5 is the fifth and current version of HTML. It has improved the markup available for documents and has introduced application programming interfaces(API) and Document Object Model(DOM).

Features:

It has introduced new multimedia features which support audio and video controls by using <audio> and <video> tags.

There are new graphics elements including vector graphics and tags.

Enrich semantic content by including <header> <footer>, <article>, <section> and <figure> are added.

Drag and Drop- The user can grab an object and drag it further dropping it to a new location.

Geo-location services- It helps to locate the geographical location of a client.

Web storage facility which provides web application methods to store data on the web browser.

Bootstrap4

Bootstrap 4 is the newest version of Bootstrap. Bootstrap can be defined as a free and open-source framework that can be used to create responsive, mobile-first, front-end web pages

Advantages of Bootstrap:

There are various different advantages of Bootstrap, a few of them are given below:

The first and foremost advantage of using Bootstrap is that it is very easy to use and implement. If a person has some basic knowledge of HTML and CSS, that user can easily use Bootstrap.

The fact that Bootstrap can adapt to the size of any phone, tablet, desktop and so on is also very interesting feature.

Bootstrap 4 is also useful because it is compatible with all modern browsers which include Google Chrome, Firefox, Internet Explorer 10+, Edge, Safari, and Opera.

It also produces less cross-browser bugs.

It is light weighted and consequently it can be widely used as a framework for creating responsive sites.

Lastly, Bootstrap 4 is a very simple and yet very effective grid system.

Why should one use Bootstrap?

There are multiple reasons for that, such as:

It is a way more fast and simpler way to develop websites.

It can create platform-independent web pages.

It can also create responsive web pages.

The web pages created by using Bootstrap 4 can be easily made responsive.

Javascript

What is JavaScript ?

JavaScript is a dynamic computer programming language. It is lightweight and most commonly used as a part of web pages, whose implementations allow client-side script to interact with the user and make dynamic pages. It is an interpreted programming language with object-oriented capabilities.

JavaScript was first known as LiveScript, but Netscape changed its name to JavaScript, possibly because of the excitement being generated by Java. JavaScript made its first appearance in Netscape 2.0 in 1995 with the name LiveScript. The general-purpose

core of the language has been embedded in Netscape, Internet Explorer, and other web browsers.

Advantages of JavaScript

The merits of using JavaScript are −

Less server interaction − You can validate user input before sending the page off to the server. This saves server traffic, which means less load on your server.

Immediate feedback to the visitors − They don't have to wait for a page reload to see if they have forgotten to enter something.

Increased interactivity − You can create interfaces that react when the user hovers over them with a mouse or activates them via the keyboard.

Richer interfaces − You can use JavaScript to include such items as drag-and-drop components and sliders to give a Rich Interface to your site visitors.

Jquery

jQuery is a fast, small, cross-platform and feature-rich JavaScript library. It is designed to simplify the client-side scripting of HTML. It makes things like HTML document traversal and manipulation, animation, event handling, and AJAX very simple with an easy-to-use API that works on a lot of different type of browsers

The main purpose of jQuery is to provide an easy way to use JavaScript on your website to make it more interactive and attractive. It is also used to add animation.

What is jQuery

jQuery is a small, light-weight and fast JavaScript library. It is cross-platform and supports different types of browsers. It is also referred as ?write less do more? because it takes a lot of common tasks that requires many lines of JavaScript code to accomplish, and binds them into methods that can be called with a single line of code whenever needed. It is also very useful to simplify a lot of the complicated things from JavaScript, like AJAX calls and DOM manipulation.

jQuery is a small, fast and lightweight JavaScript library.

jQuery is platform-independent.

jQuery means "write less do more".

jQuery simplifies AJAX call and DOM manipulation.

Thymeleaf

The Thymeleaf is an open-source Java library that is licensed under the Apache License 2.0. It is a HTML5/XHTML/XML template engine. It is a server-side Java template engine for both web (servlet-based) and non-web (offline) environments. It is perfect for modern-day HTML5 JVM web development. It provides full integration with Spring Framework.

It applies a set of transformations to template files in order to display data or text produced by the application. It is appropriate for serving XHTML/HTML5 in web applications.

The goal of Thymeleaf is to provide a stylish and well-formed way of creating templates. It is based on XML tags and attributes. These XML tags define the execution of predefined logic on the DOM (Document Object Model) instead of explicitly writing that logic as code inside the template. It is a substitute for JSP.

The architecture of Thymeleaf allows the fast processing of templates that depends on the caching of parsed files. It uses the least possible amount of I/O operations during execution.

Why we use Thymeleaf?

JSP is more or less similar to HTML. But it is not completely compatible with HTML like Thymeleaf. We can open and display a Thymeleaf template file normally in the browser while the JSP file does not.

Thymeleaf supports variable expressions (${...}) like Spring EL and executes on model attributes, asterisk expressions (*{...}) execute on the form backing bean, hash expressions (#{...}) are for internationalization, and link expressions (@{...}) rewrite URLs.

Like JSP, Thymeleaf works well for Rich HTML emails.

Spring framework

Spring is the most popular application development framework for enterprise Java. Millions of developers around the world use Spring Framework to create high performing, easily testable, and reusable code.

Spring framework is an open source Java platform. It was initially written by Rod Johnson and was first released under the Apache 2.0 license in June 2003.

Spring is lightweight when it comes to size and transparency. The basic version of Spring framework is around 2MB.

The core features of the Spring Framework can be used in developing any Java application, but there are extensions for building web applications on top of the Java EE platform. Spring framework targets to make J2EE development easier to use and promotes good programming practices by enabling a POJO-based programming model.

Benefits of Using the Spring Framework

Following is the list of few of the great benefits of using Spring Framework −

Spring enables developers to develop enterprise-class applications using POJOs. The benefit of using only POJOs is that you do not need an EJB container product such as an application server but you have the option of using only a robust servlet container such as Tomcat or some commercial product.

Spring is organized in a modular fashion. Even though the number of packages and classes are substantial, you have to worry only about the ones you need and ignore the rest.

Spring does not reinvent the wheel, instead it truly makes use of some of the existing technologies like several ORM frameworks, logging frameworks, JEE, Quartz and JDK timers, and other view technologies.

Testing an application written with Spring is simple because environment-dependent code is moved into this framework. Furthermore, by using JavaBeanstyle POJOs, it becomes easier to use dependency injection for injecting test data.

Spring's web framework is a well-designed web MVC framework, which provides a great alternative to web frameworks such as Struts or other over-engineered or less

popular web frameworks.

Spring provides a convenient API to translate technology-specific exceptions (thrown by JDBC, Hibernate, or JDO, for example) into consistent, unchecked exceptions.

Lightweight IoC containers tend to be lightweight, especially when compared to EJB containers, for example. This is beneficial for developing and deploying applications on computers with limited memory and CPU resources

Spring provides a consistent transaction management interface that can scale down to a local transaction (using a single database, for example) and scale up to global transactions (using JTA, for example).

Spring boot

Spring Boot is an open source Java-based framework used to create a micro Service. It is developed by Pivotal Team and is used to build stand-alone and production ready spring applications. This chapter will give you an introduction to Spring Boot and familiarizes you with its basic concepts.

What is Micro Service?

Micro Service is an architecture that allows the developers to develop and deploy services independently. Each service running has its own process and this achieves the lightweight model to support business applications.

Advantages

Micro services offers the following advantages to its developers −

Easy deployment

Simple scalability

Compatible with Containers

Minimum configuration

Lesser production time

What is Spring Boot?

Spring Boot provides a good platform for Java developers to develop a stand-alone and production-grade spring application that you can just run. You can get started with minimum configurations without the need for an entire Spring configuration setup.

Advantages

Spring Boot offers the following advantages to its developers −

Easy to understand and develop spring applications

Increases productivity

Reduces the development time

Goals

Spring Boot is designed with the following goals −

To avoid complex XML configuration in Spring

To develop a production ready Spring applications in an easier way

To reduce the development time and run the application independently

Offer an easier way of getting started with the application

Spring security

Spring Security is a framework which provides various security features like: authentication, authorization to create secure Java Enterprise Applications.

It is a sub-project of Spring framework which was started in 2003 by Ben Alex. Later on, in 2004, It was released under the Apache License as Spring Security 2.0.0.

It overcomes all the problems that come during creating non spring security applications and manage new server environment for the application.

This framework targets two major areas of application are authentication and authorization. Authentication is the process of knowing and identifying the user that wants to access.

Authorization is the process to allow authority to perform actions in the application.

We can apply authorization to authorize web request, methods and access to individual domain.

Advantages

Spring Security has numerous advantages. Some of that are given below.

Comprehensive support for authentication and authorization.

Protection against common tasks

Servlet API integration

Integration with Spring MVC

Portability

CSRF protection

Java Configuration support

Spring data jpa with hibernate

Hibernate is a Java framework that simplifies the development of Java application to interact with the database. It is an open source, lightweight, ORM (Object Relational Mapping) tool. Hibernate implements the specifications of JPA (Java Persistence API) for data persistence.

ORM Tool

An ORM tool simplifies the data creation, data manipulation and data access. It is a programming technique that maps the object to the data stored in the database.

What is JPA?

Java Persistence API (JPA) is a Java specification that provides certain functionality and standard to ORM tools. The javax.persistence package contains the JPA classes and interfaces.

Advantages of Hibernate Framework

Following are the advantages of hibernate framework:

1) Open Source and Lightweight

Hibernate framework is open source under the LGPL license and lightweight.

2) Fast Performance

The performance of hibernate framework is fast because cache is internally used in hibernate framework. There are two types of cache in hibernate framework first level cache and second level cache. First level cache is enabled by default.

3) Database Independent Query

HQL (Hibernate Query Language) is the object-oriented version of SQL. It generates the database independent queries. So you don't need to write database specific queries. Before Hibernate, if database is changed for the project, we need to change the SQL query as well that leads to the maintenance problem.

4) Automatic Table Creation

Hibernate framework provides the facility to create the tables of the database automatically. So there is no need to create tables in the database manually.

5) Simplifies Complex Join

Fetching data from multiple tables is easy in hibernate framework.

6) Provides Query Statistics and Database Status

Hibernate supports Query cache and provide statistics about query and database status.

Chapter 2
Proposed System

2.1 Proposed System:

• The proposed system help in building application to buy, sell product or good online using internet connection.

• Ecommerce is fast gaining ground as an accepted and used business paradigm.

• It provides searching facility by the product name so customers can easily search the products accordingly.

• In proposed system every product description is given along the product.

• System will be error free, secure, reliable and fast management system.

• In this system we make user friendly environment to the user and provides the continuous availability of the products.

• Website provide cash on delivery service.

2.2 Objectives of System:

• To provide every product for the customer in the minimum reasonable cost, customer can purchase product every time.

• To manage details of products, category, brand, customer, shipping rates

• The purpose of System is to automate the existing manual system by the help of computerized and full-fledged computer website, fulfilling their requirements, so that their valuable data/information can be stored for a longer period.

• To storing the high amount of data in minimum time is less time consuming for the organization.

• It is designed such a way that one can view all the updates of the products from any place through online.

2.3 User Requirements:

➢ **Requirement for Admin:**

• Registration

• Login

• Add/Manage User

• Add/Manage Product

• Add/Manage category

• Manage Customers

• Add/Manage Shipping Rates

• View Orders

➢ **Requirement for Customer:**

• Registration

• Login

• View Product

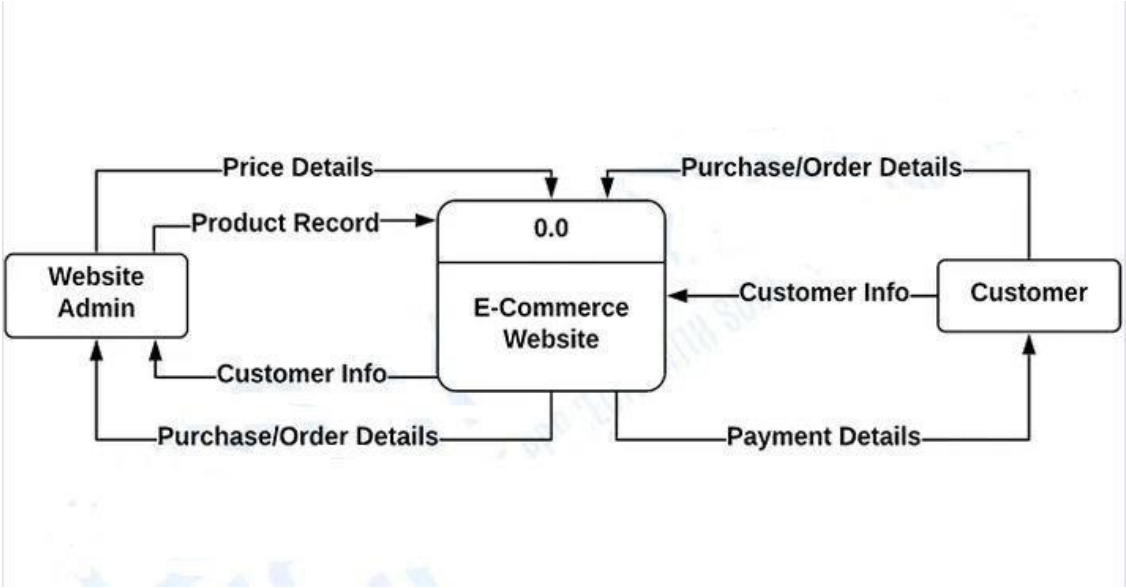
• Search facility

• Review Product

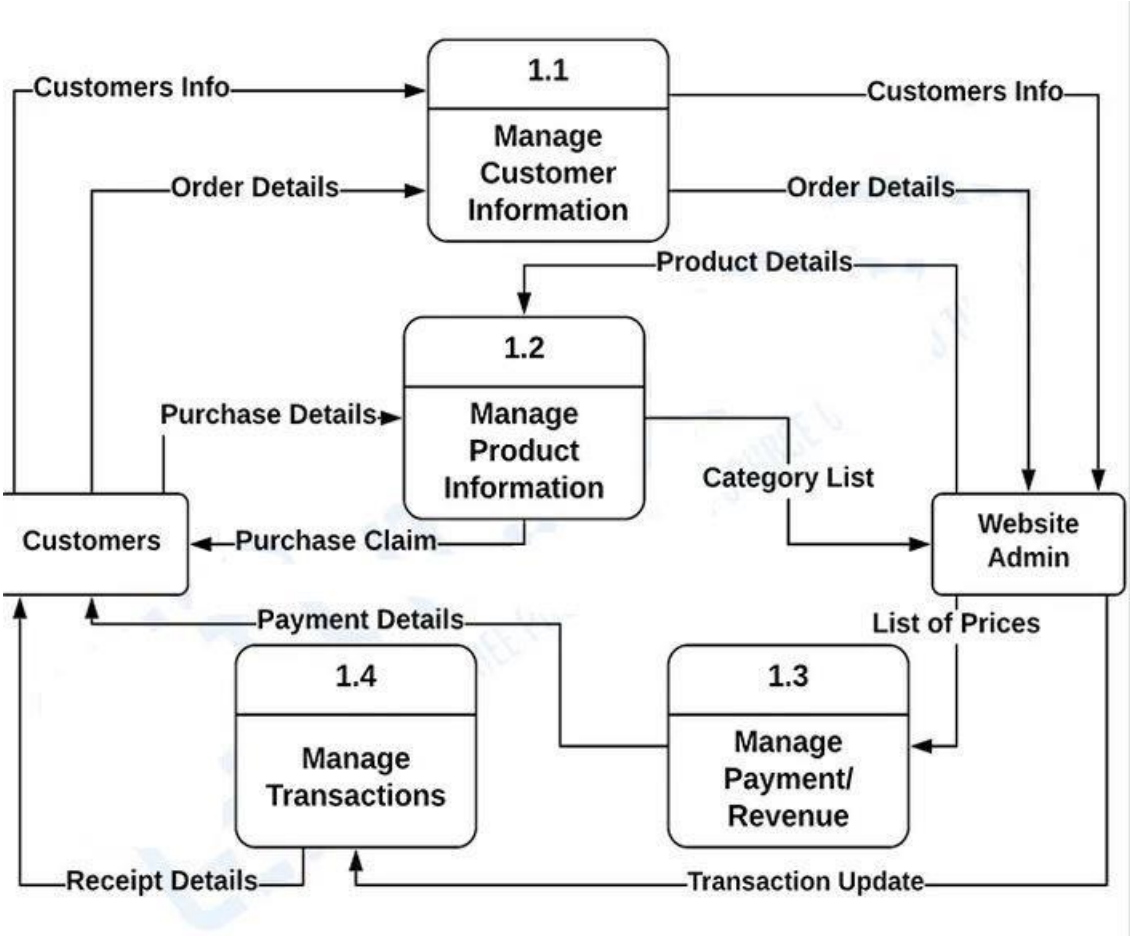
• Order Product

• Change password

3.1.1 DFD Diagram:



3.1.2 FIRST LEVEL DFD



3.2 Table specifications (Database) :

Data Dictionary:

Table Name: Address

Table Name	Address	
Table Description	It stores Address details.	
Primary_key	Address_id	
Foreign Key	custoId, countryId	
Field Name	**Data type**	**Description**
First_name	Varchar(45)	Store first name
Last_name	Varchar(45)	Store last name
Phone_number	Varchar(45)	Store phone number
Address_line_1	Varchar(45)	Store address line
Address_line_2	Varchar(45)	Store address line
city	Varchar(45)	Store city code
state	Varchar(45)	Store state details
Postal code	Varchar(45)	Store postal code

Table Name: Customers

Table Name	Customers	
Table Description	It stores customers details.	
Primary_key	Cust_id	
Foreign Key	Country_id	
Field Name	**Data type**	**Description**
Email	Varchar(45)	Store email id of customer
Password	Varchar(45)	Store password of customer
First_name	Varchar(45)	Store first name of customer
Last_name	Varchar(45)	Store last name of customer
Phone_number	Varchar(45)	Store phone number of customer
Address_line_1	Varchar(45)	Store address of customer
Address_line_2	Varchar(45)	Store address of customer
City	Varchar(45)	Store city of customer
State	Varchar(45)	Store state of customer
Postal code	Varchar(45)	Store postal code of customer

Table Name: Orders

Table Name	Orders	
Table Description	It stores order details.	
Primary_key	Order_id	
Foreign Key	custId	
Field Name	**Data type**	**Description**
Order_time	int(10)	Store bill id.
Payment_method	varchar(255)	Store bill date.
Product_cost	Float	Bill amount.
Shipping_cost	Float	Store shipping cost
Tax	Float	Store tax
Total	Float	Store total
First Name	Varchar(45)	Store firstname
Last Name	Varchar(45)	Store lastname
Phone_number	Varchar(15)	Store phonenumber
Address_line_1	Varchar(45)	Store address
City	Varchar(45)	Store city

Table Name: Shipping Rates

Table Name	Shipping Rates	
Table Description	It stores shipping detail	
Primary_key	Id	
Foreign Key	Country_id	
Field Name	**Data type**	**Description**
State	Int	Store bill id.
Rate	Int	Store bill date.
days	Varchar(45)	Bill amount.

Table Name: Review

Table Name	Review	
Table Description	It stores Review details.	
Primary_key	Review_id	
Foreign Key	ProID, custID	
Field Name	**Data type**	**Description**
Comment	Varchar(300)	Store comments
headline	Varchar(128)	Store headline
Rating	Varchar(200)	Store rating
Review time	DateTime(6)	Store review time

Table Name: Order Details

Table Name	Order_Details	
Table Description	Order details	
Primary_key	Id	
Foreign Key	Order_id, product_id	
Field Name	**Data type**	**Description**
Quantity	int(10)	Store Quantity id.
Unit_Price	Float(20)	Store Unit Price.
Subtotal	int(10)	Store Subtotal.
Product_cost	float(20)	Store Product cost
Shipping_cost	float(20)	Store Shipping cost

Brands:-

Table Name	Data Type	Description.
Id	Int(10)	Store brand id
Name	Varchar(45)	Store brand name
Logo	Varchar(45)	Store brand logo

Table Name: Product

Table Name	Product	
Table Description	It stores product details.	
Primary_key	Id	
Foreign Key	Brand_id, category_id	
Field Name	**Data type**	**Description**
Name	Int	Store bill id.
Alias	Varchar(256)	Store bill date.
Short_description	Varchar(256)	Bill amount.
Full_description	Varchar(512)	Store description of product
Main_page	Varchar(255)	Store main page of product
Created_time	datetime(6)	Store created time
Updated_time	Datetime(6)	Store updates time
Enabled	Boolean	Status enabled or not
In_stocks	Boolean	Store stock
cost	Float	Store cost of product
price	Float	Store price of product

Table Name: Category

Table Name	Category	
Table Description	It stores Category details.	
Primary_key	categoryId	
Foreign Key	parentId, all_parent_ids	
Field Name	**Data type**	**Description**
Name	String(15)	Name of the Category
Alias	int(10)	Alias of Category
Image	Varchar(128)	Image of category
Enabled	Varchar(64)	Enabled

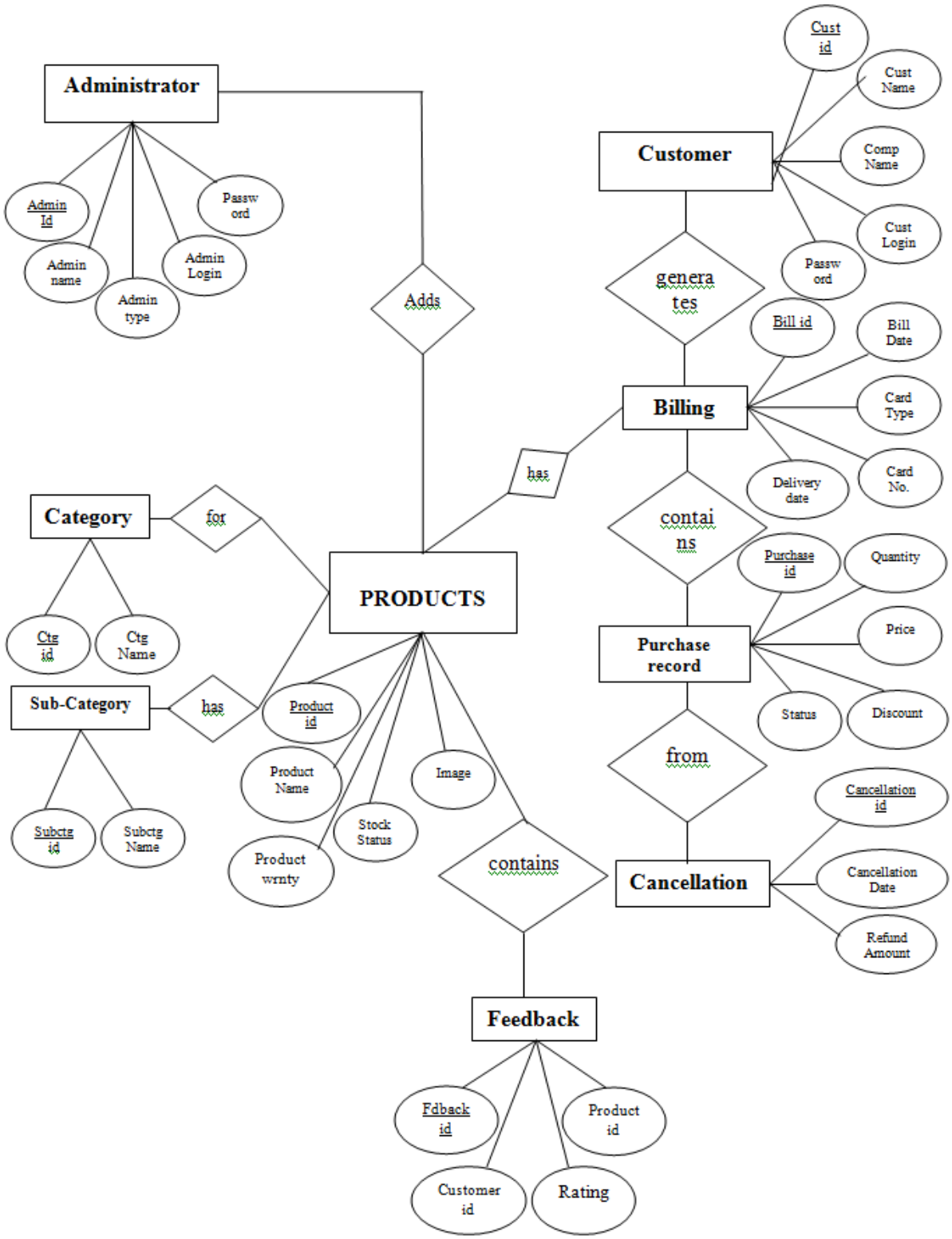
Table Name: Order_Track

Table Name	Order_Track	
Table Description	It stores Order Track.	
Primary_key	Id	
Foreign Key	Order_id	
Field Name	**Data type**	**Description**
Order_id	int(10)	Stores Id of order
Status	Varchar(10)	Stores Status of order
Updated_time	Varchar(10)	Stores Updated time

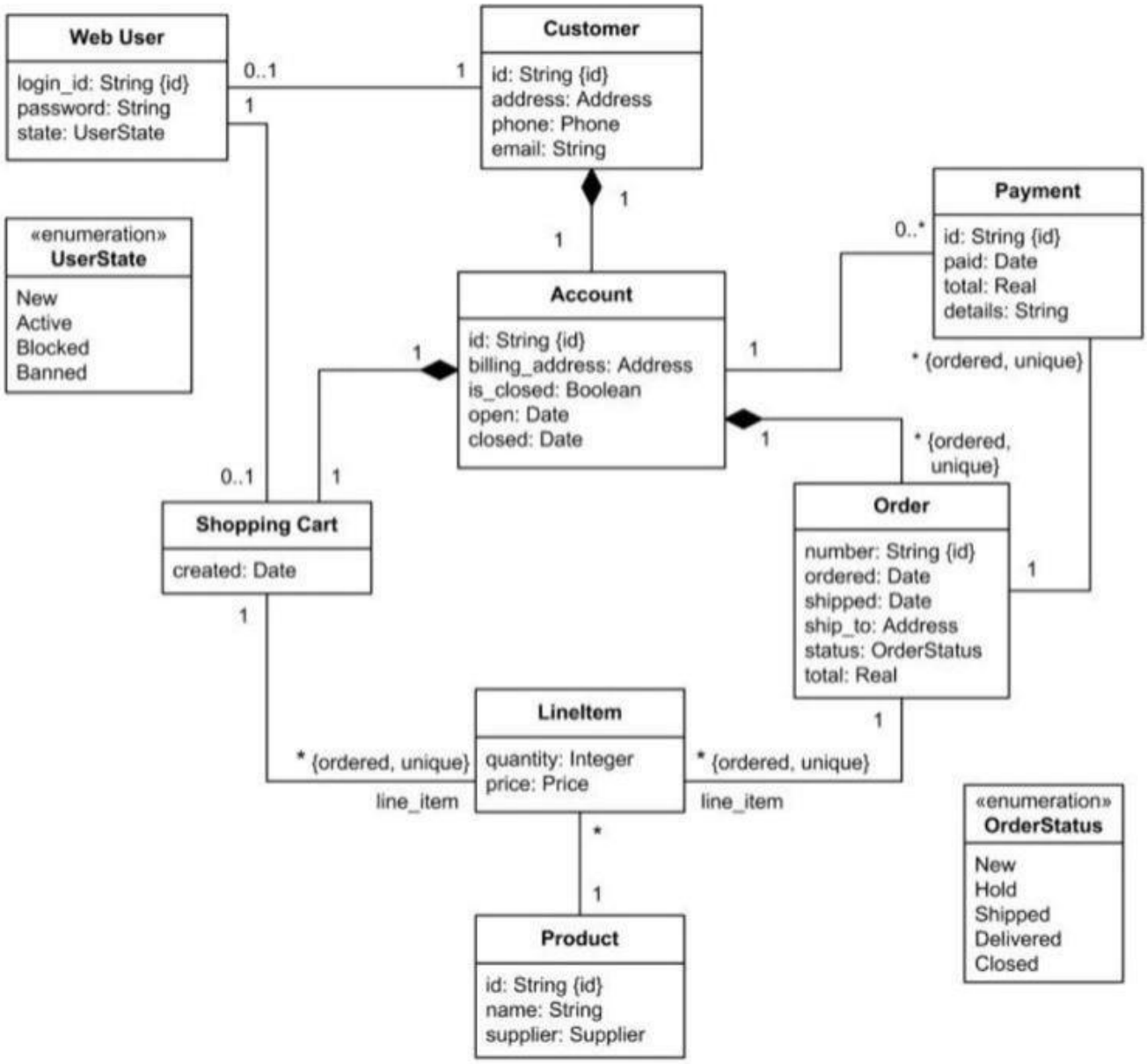
Table Name: Users

Table Name	Users	
Table Description	It stores Users details	
Primary_key	user_id	
Field Name	**Data type**	**Description**
Email	Varchar(128)	Stores email
Enabled	Boolean	Status shows user is enabled or not
First_name	Varchar(45)	Stores first name.
Last_name	Varchar(45)	Stores last name
password	Varchar(45)	Store password
photos	Varchar(45)	Store photos

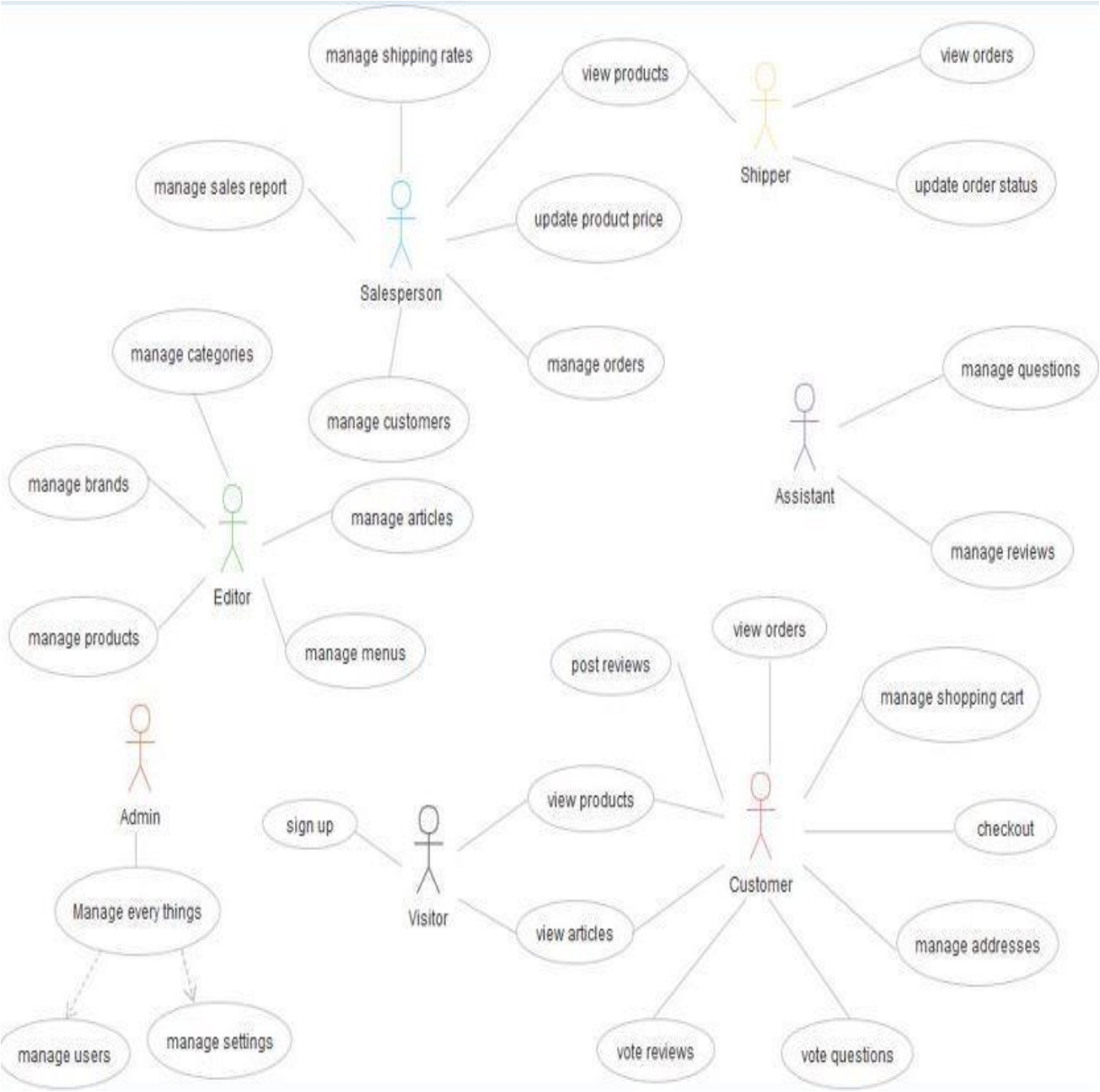
3.3 ERD Diagram:



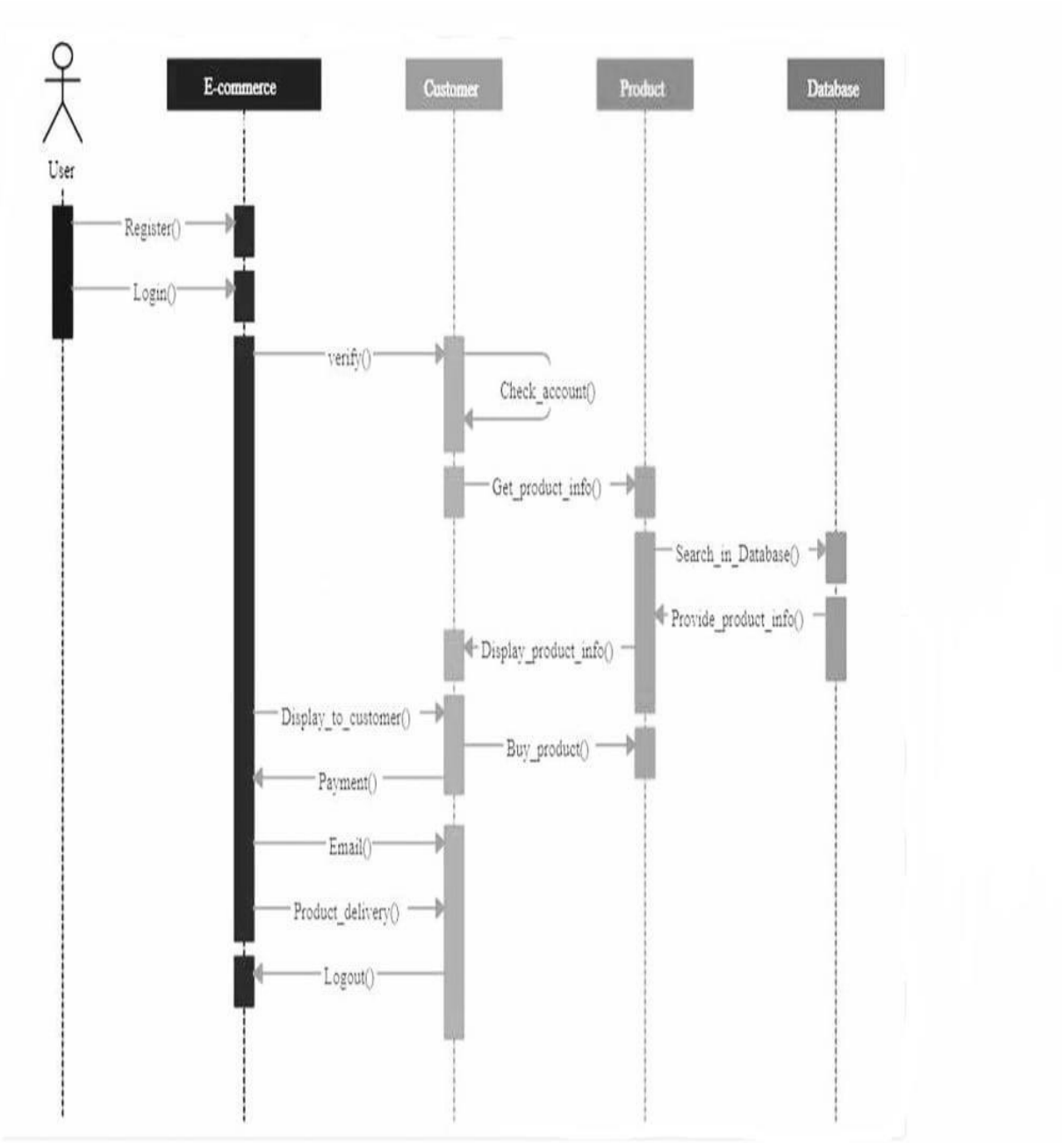
3.5 Class Diagram:



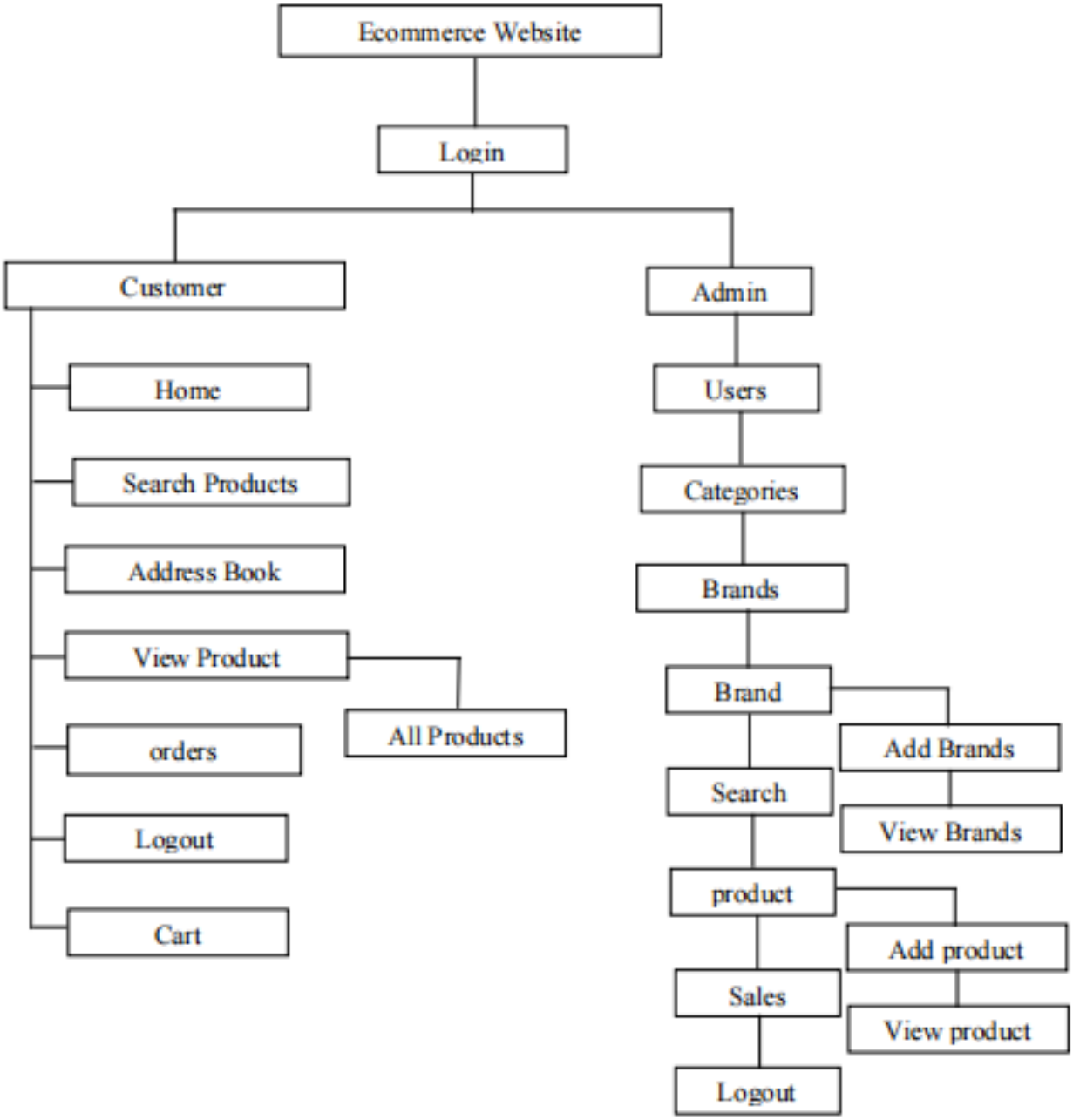
3.6 Use Case Diagrams:



3.7 sequence diagram:



3.8 Website map diagram:



CHAPTER 4:

USER MANUAL

4.1 User Interface Design(Screens etc)

Fig 1 : Admin Login Page:



Fig 2: Create New Category:

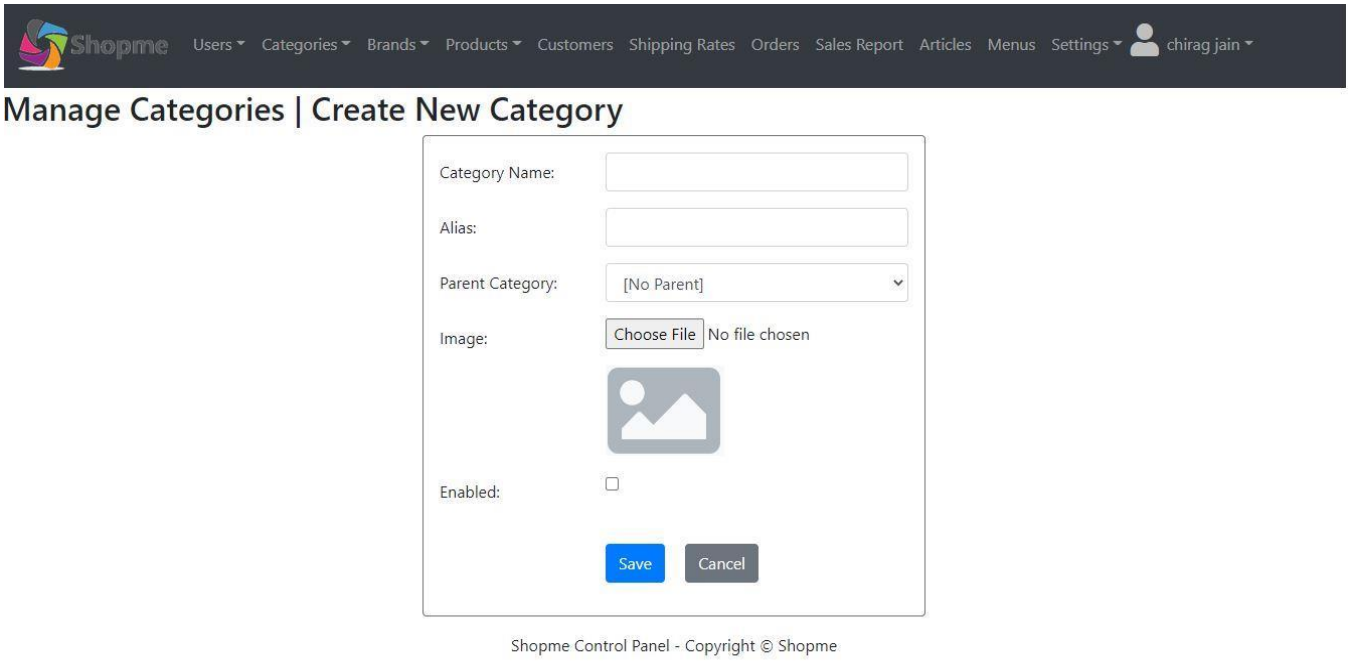


Fig 3: create new Brand

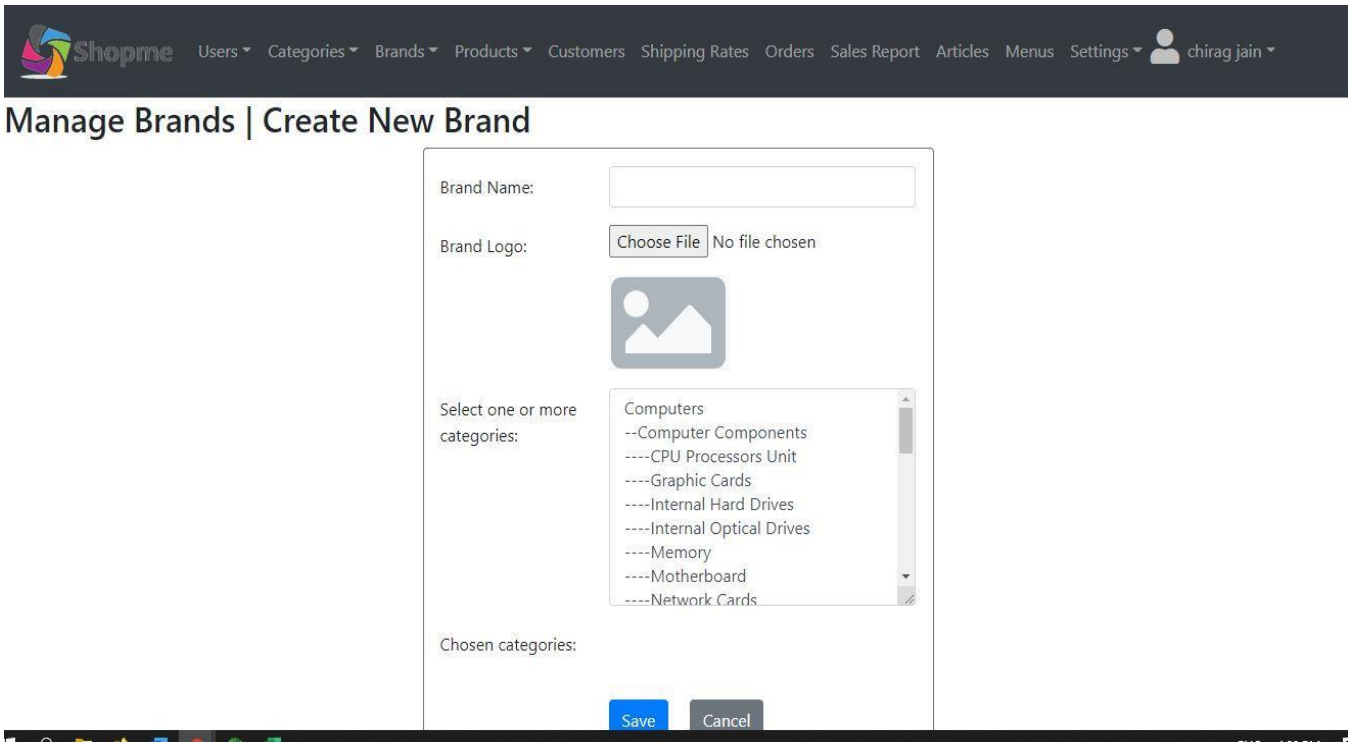


Fig 4: Manage Brands

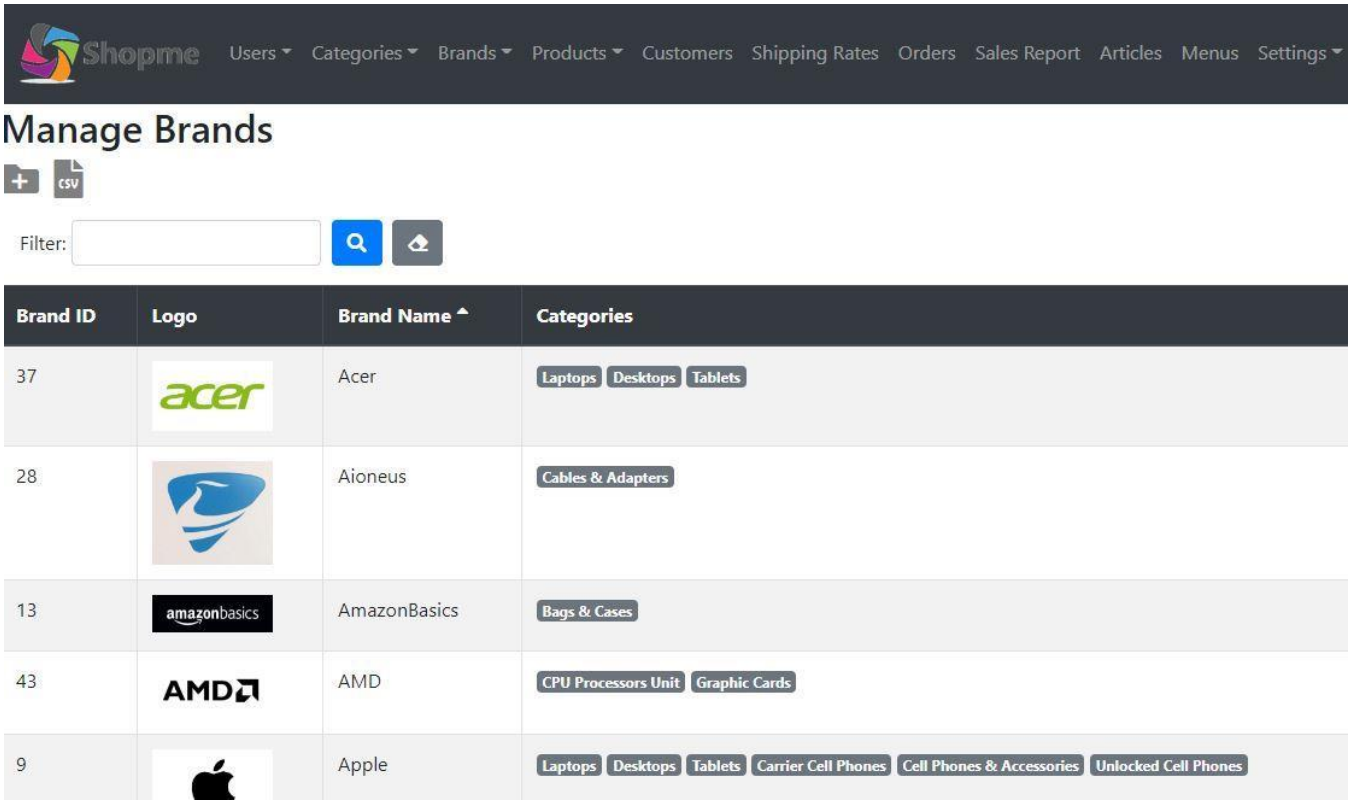
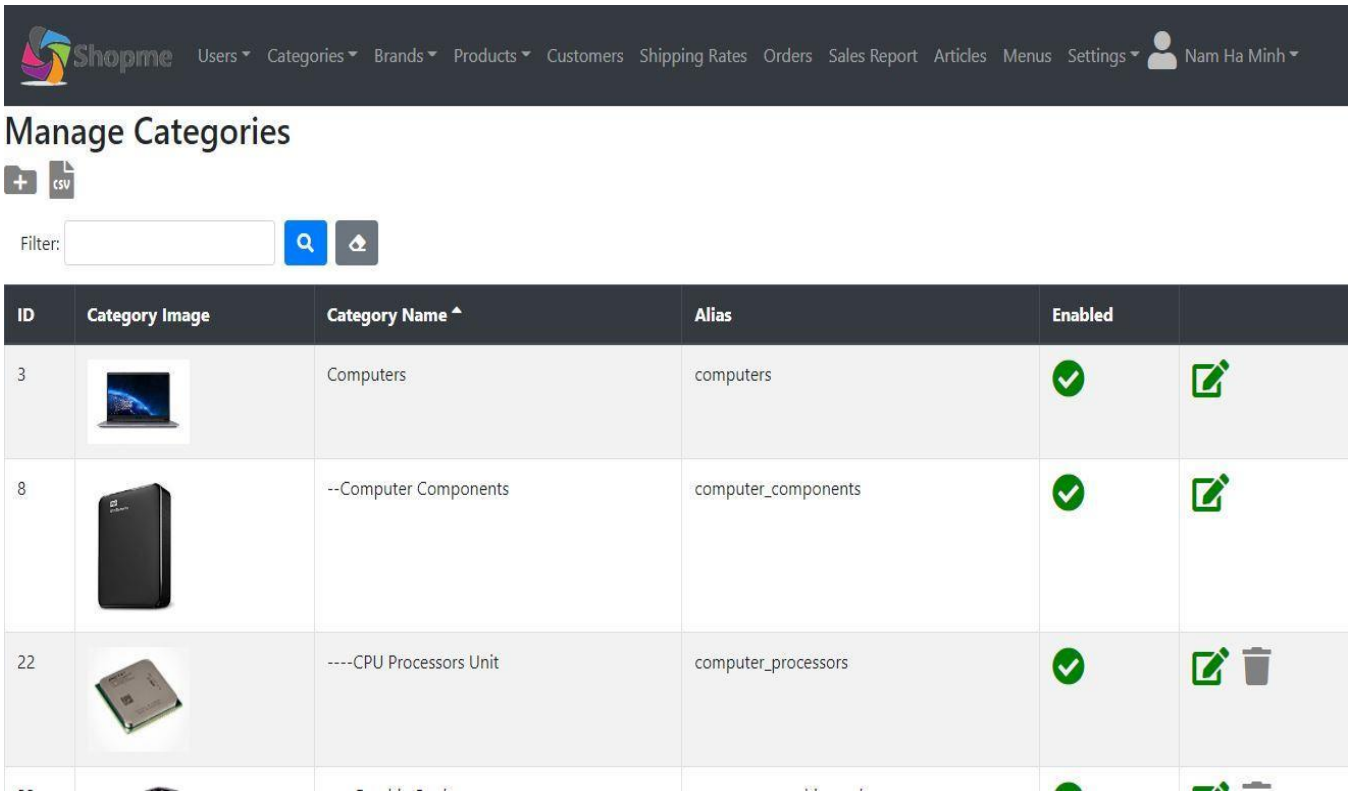


Fig 5: Manage Categories:



bb

Fig 6: Manage Customers

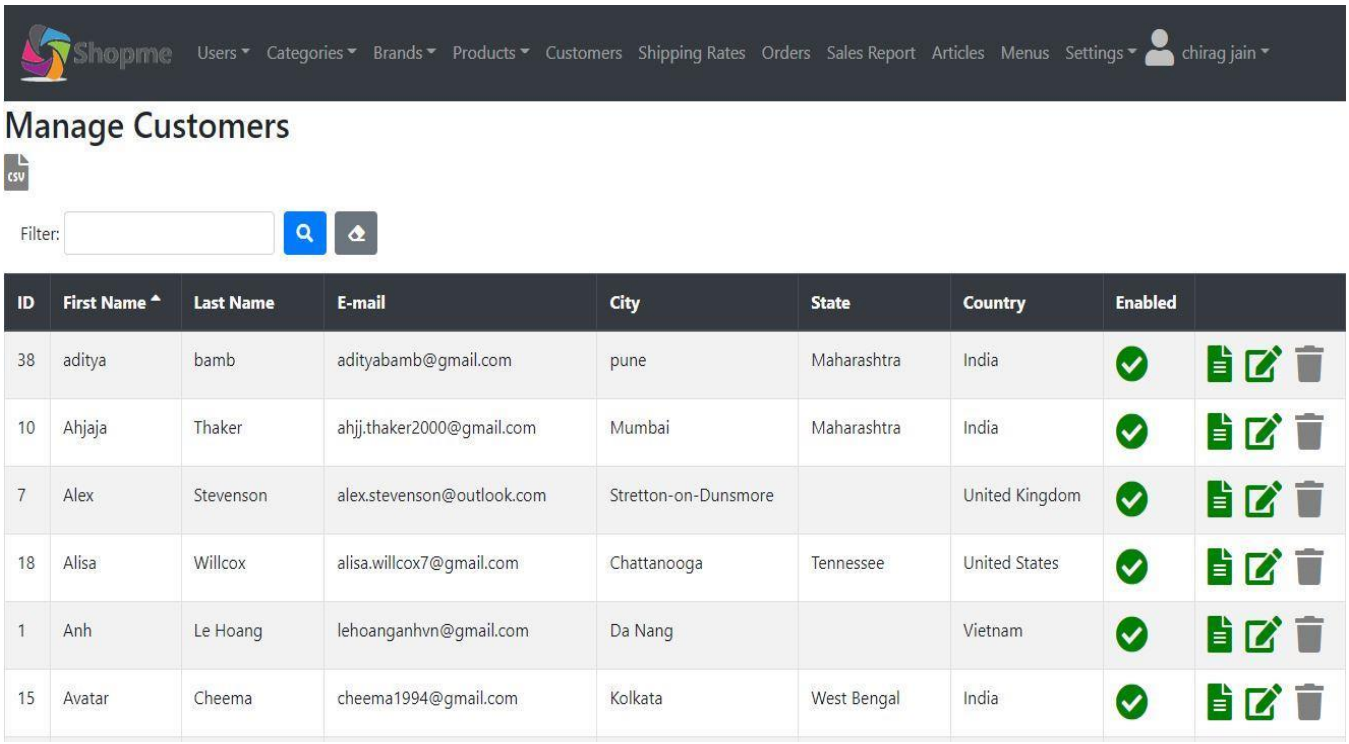


Fig 7: Manage Reviews

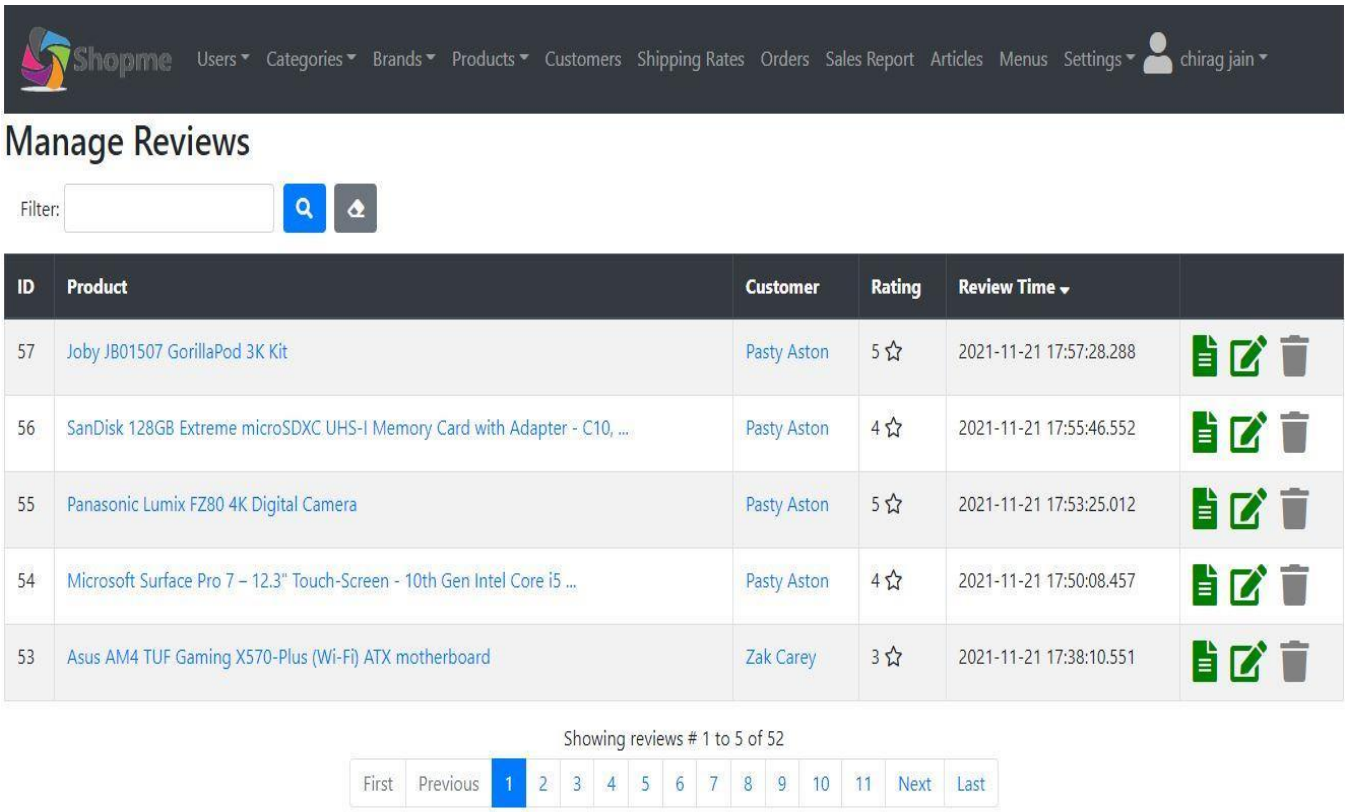


Fig 8: Manage Shipping Rates

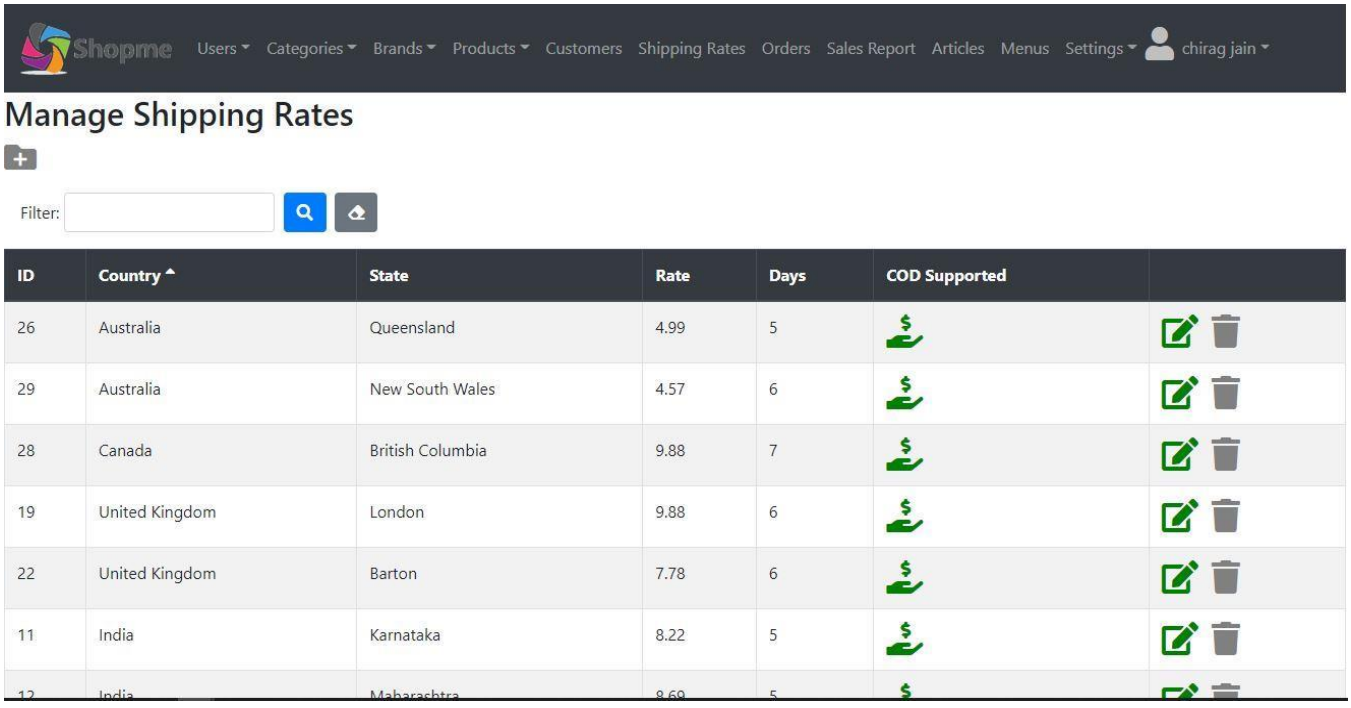


Fig 9: My Orders

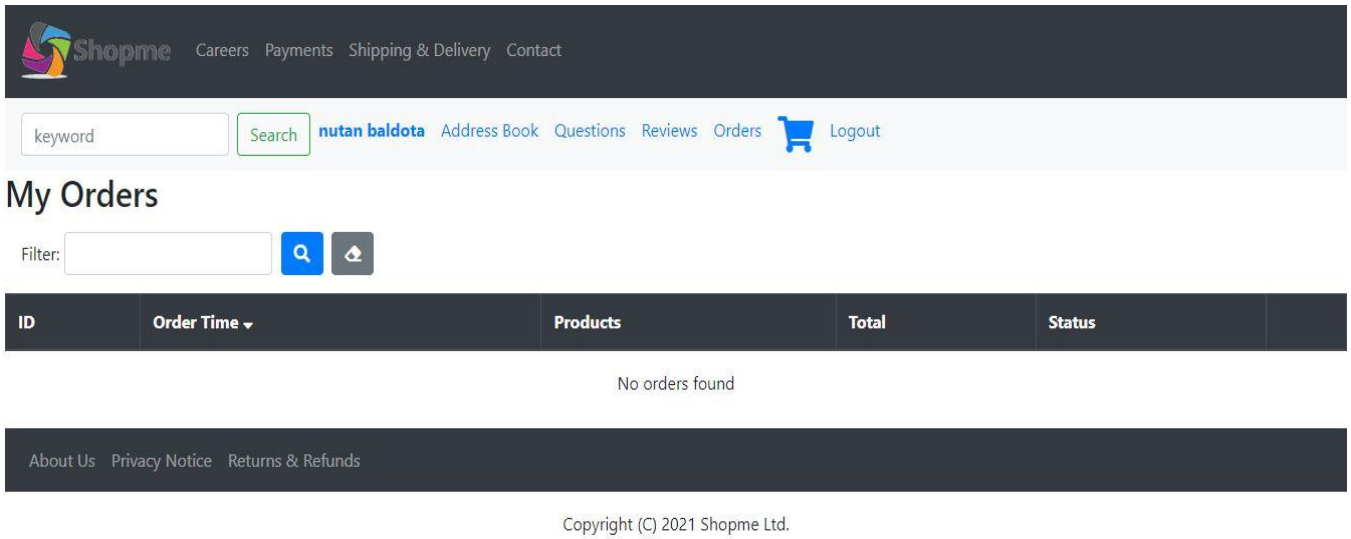


Fig 10: Sales Report

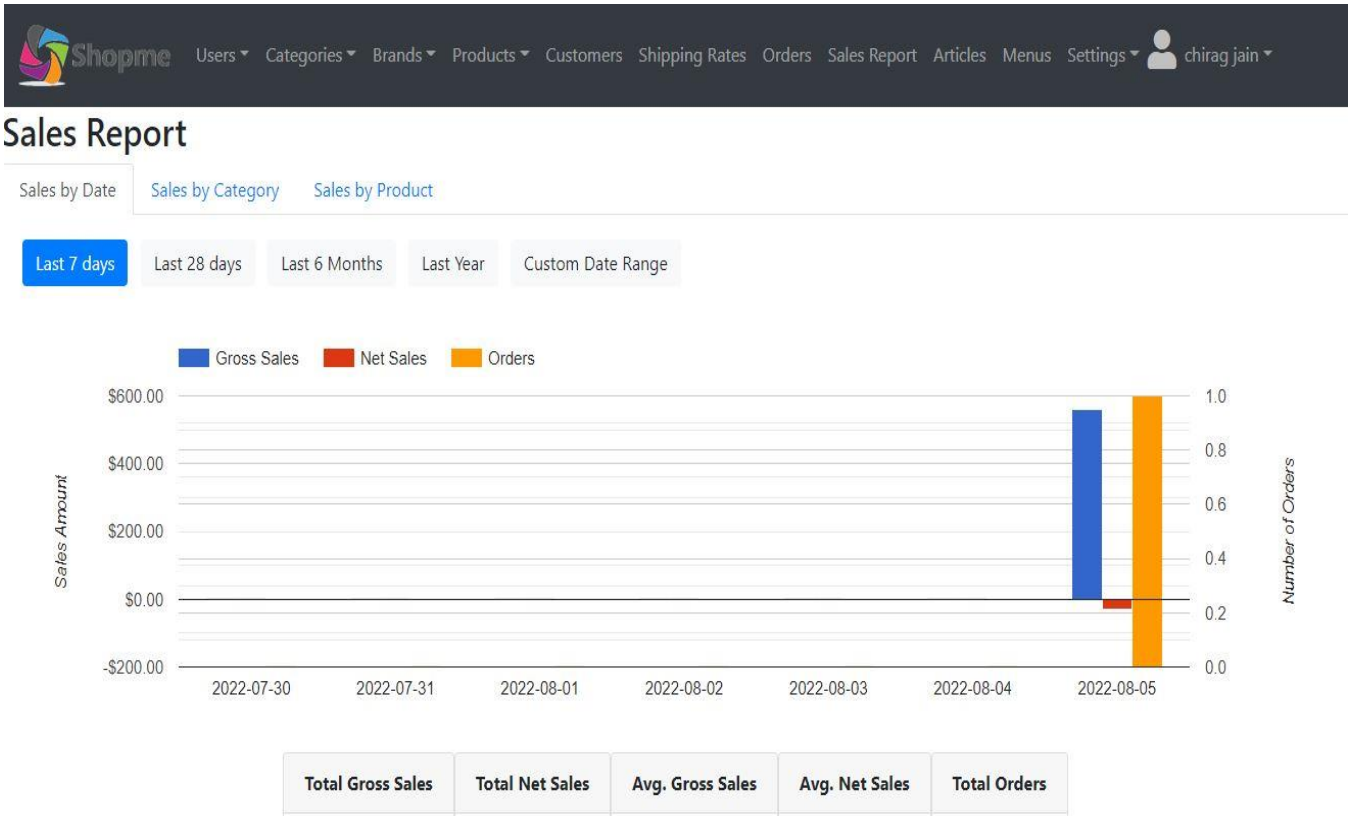
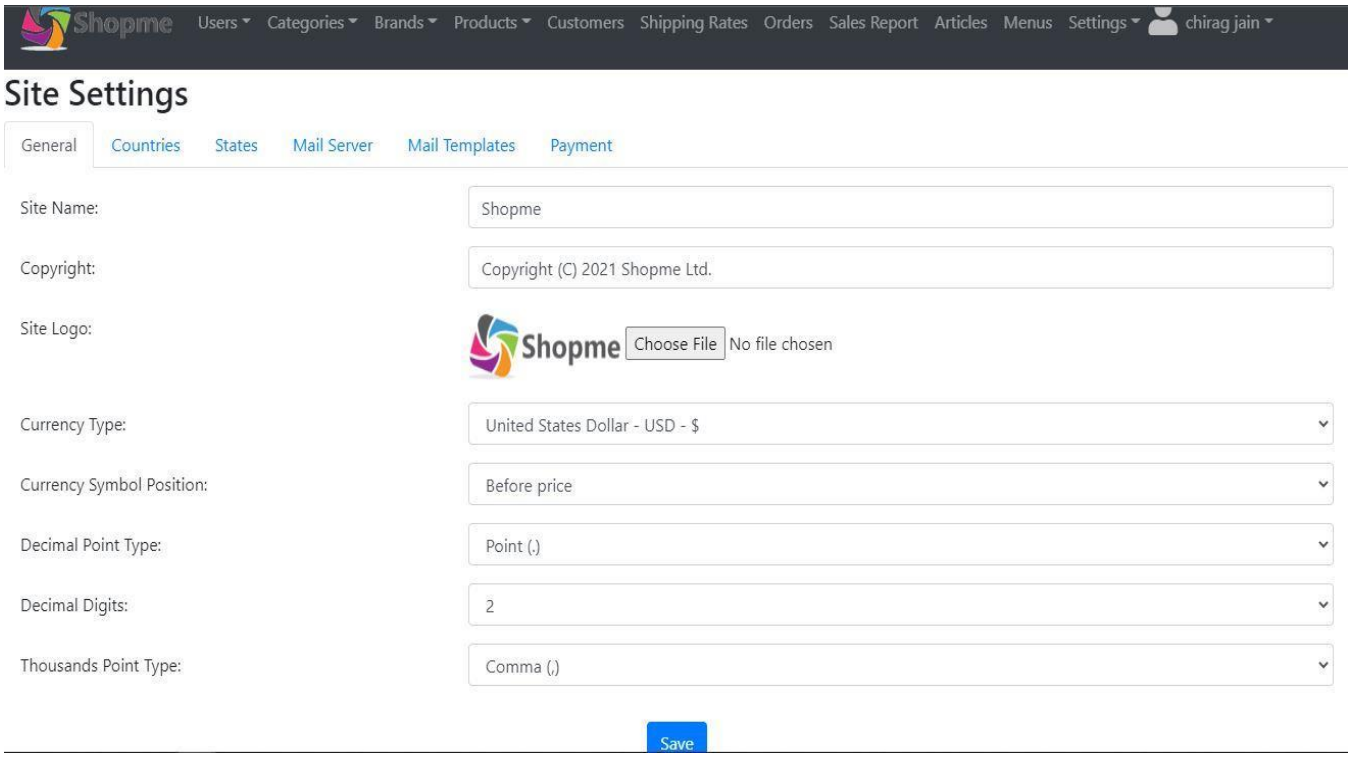


Fig 11: Site Setting



Test Cases

Test Scenario ID	Test Scenario Description	Test Case ID	Test Case Description	Test Steps	Pre-conditions	Post-conditions	Expected Result	Actual Result
TS_CRS_01	Verify the login functionality of Admin	TC_CRS_Login_01_1	Enter a valid username and password	1. Enter valid username 2. Enter passwood	Enter valid URL and Test Data	User should be able to see Dashboard	Dashboard, 1)Add Image 2)Add Menu 3)Add pages	Dashboard, 1)Add Image 2)Add Menu
		TC_CRS_Login_01_2	Enter invalid Username or password	1.Enter invalid username and valid passwood 2.Enter valid username and invalid passwood	Enter valid URL and Test Data	User should be able to see Dashboard	Dashboard	Dashboard
TS_CRS_02	Verify the login functionality of website	TC_CRS_Login_02_1	Enter a valid username and password	1. Enter valid username 2. Enter passwood	Enter valid URL and Test Data	User should be able to see Dashboard	1.View wp dashboard 2. Logout button at right top side of page	1.View wp dashboard 2. Logout button at right top side of page
		TC_CRS_Login_02_2	Enter invalid Username or password	1.Enter invalid username and valid passwood 2.Enter valid username and invalid passwood	Enter valid URL and Test Data	User should be able to see home page	Reset password	Reset password
TS_CRS_03	Verify the product functionality of website	TC_CRS_Login_03_1	Enter a valid product name and price	1. Enter Valid name 2. Enter valid price	Enter valid URL and product data	User should be able to see home page	1.view product on home page	1.view product on home page
			Enter a valid product name and price	1.Enter invalid product name and valid price 2.Enter valid price	Enter valid URL and product data	User should be able to see home page	Homepage	Homepage
TS_CRS_04	Verify the functionalities of Add users	TS_CRS Add_04_1	1.Enter First name	1.First letter should be capital 2. Only First name should be added	Enter valid username and password of Admin	User should be able to see Add page view	1.First name should be displayed	1.First name should be displayed

Test Scenario ID	Test Scenario Description	Test Case ID	Test Case Description	Test Steps	Pre-conditions	Post-conditions	Expected Result	Actual Result
		TS_CRS Add_04_2	1.Enter Other name	1.First letter should be capital 2. Only Other names name should be added	Enter valid username and password of Admin	User should be able to see Add page view	1.Other name should be displayed	1.Other name should be displayed
		TS_CRS Add_04_3	1.Enter users Number	1. User number should be less than 5 digits	Enter valid username and password of Admin	User should be able to see Add page view	1.member number should be displayed	1.member number should be displayed
		TS_CRS Add_04_4	1.Enter Password	1. Passwords should be more than 7 characters. 2. Password should contain at least one digit and special character	Enter valid username and password of Admin	User should be able to see Add page view	1.Password should be in encrypted form	1. Password should be in encrypted form
		TS_CRS Add_04_5	1.Select status	1. Select Status should contain all the status. 2. Status should not be empty	Enter valid username and password of Admin	User should be able to see Add page view	1.Status should select only one option	1.Status should select only one option
TS_CRS _05	Verify the functionality of View products	TC_CRS_View_05_1	Search button should be enabled	1.Search button should be enabled 2. User should be able to type	product list should be displayed	User should be able to see the list of added products	Entered name in search tab should be available	Entered name in search tab should be available
		TC_CRS_View_05_2	Edit popup should be displayed	1.Edit button should be enabled 2. Edit popup displayed 3. All fields must be in editable mode	product list must display edit option	Edit pop up should be displayed	All fields of edit pop up must be in editable mode	All fields of edit pop up must be in editable mode
		TC_CRS_View_05_3	Delete button should be displayed	1.Delete button should be enabled 2. Delete popup should be displayed 3.product must be published	User list must display delete option	Display pop up should be displayed	Delete pop up must display product and delete button should be enabled	Delete pop up must display product and delete button should be enabled

Test Scenario ID	Test Scenario Description	Test Case ID	Test Case Description	Test Steps	Pre-conditions	Post-conditions	Expected Result	Actual Result
TS_CRS_07	Verify the functionality of add pages	TC_CRS_add_pages_07_1	Edit popup should be displayed	1.Edit button should be enabled 2. Edit popup displayed 3. All fields must be in editable mode	pages list must display edit option	Edit pop up should be displayed	All fields of edit pop up must be in editable mode	All fields of edit pop up must be in editable mode
		TC_CRS_assign_case_07_2	Details button should be enabled	1.Print button should be present 2. page details must be present 3.description detail must be present	page must be assigned	User must be able to see the pages	Print button must be present Report must be generated of sales	Print button must be present Report must be generated of sales
TS_CRS_08	Verify the functionalities of new order	TS_CRS_new_case_08_1	1.Enter Name of customer	1.name address and billing details should be added	Enter valid username and password of user	User should be able to see add new product page	1.Name of users order should be display	1.Name of users order should be display
		TS_CRS_new_case_08_2	1.Enter Phone No	1. Phone number should be less than 11 digits	Enter valid username and password of new	User should be able to see add new order	1.Phone Number should be displayed	1.Phone Number should be displayed
		TS_CRS_new_case_08_3	1.Enter billing address	1.Occupation should accept only letters	Enter valid username and password of NCO	User should be able to see add new order	1.Occupation should be displayed	1.Occupation should be displayed
		TS_CRS_new_case_08_4	1.Enter mail id	1.Mail id should contain @ into it	Enter valid username and password of NCO	User should be able to see add new order	1..Address should be displayed	1..Address should be displayed
		TS_CRS_new_case_08_5	1.Select Gender	1. Select Gender should contain all the options.	Enter valid username and password of NCO	User should be able to see add new order	1.Gender should be displayed	1.Gender should be displayed
		TS_CRS_new_case_08_6	1.Enter shipping address	1.Address should contain city, house number, landmark, region, district	Enter valid username and password of NCO	User should be able to see add new case page view	1..Address should be displayed	1..Address should be displayed
		TS_CRS_new_case_08_7	1.Enter payment type	1. Payment Type should contain COD and credit card	Enter valid username and password of NCO	User should be able to see add new order	1.Payment Type should be displayed	1.Payment Type should be displayed

4.2 Limitations

1. Some business process may never be available to ecommerce:- Some items such as foods, high cost items such as jwellery may be impossible to be available on the internet.

2. Legal issues:- these issues arises when the customer data is fall in the hands of strangers.

3. Negotiation is not possible on online platform.

4,3 Future Enhancement

• Product recommendation using machine learning algorithm .

• Smart chat bot to improve customer service.

4,4 Bibliography

Website

• www.google.com

• https://spring.io/learn

• www.javapoint.com

Books

1. Spring microservices in Action

Author - john cornel

Publisher – Manning, July 6 2017

2 Learning Spring boot

Author – greg turnquest

Publisher – packt, November 3, 2017

ANNEXURE: SAMPLE PROGRAM CODE

BrandController.java –

```
package com.shopme.admin.brand;

import java.io.IOException;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;

import org.springframework.stereotype.Controller;

import org.springframework.ui.Model;

import org.springframework.util.StringUtils;

import org.springframework.web.bind.annotation.GetMapping;

import org.springframework.web.bind.annotation.PathVariable;

import org.springframework.web.bind.annotation.PostMapping;

import org.springframework.web.bind.annotation.RequestParam;

import org.springframework.web.multipart.MultipartFile;

import org.springframework.web.servlet.mvc.support.RedirectAttributes;

import com.shopme.admin.AmazonS3Util;

import com.shopme.admin.category.CategoryService;

import com.shopme.admin.paging.PagingAndSortingHelper;

import com.shopme.admin.paging.PagingAndSortingParam;

import com.shopme.common.entity.Brand;

import com.shopme.common.entity.Category;

@Controller

public class BrandController {

private String defaultRedirectURL =

"redirect:/brands/page/1?sortField=name&sortDir=asc";

@Autowired private BrandService brandService;

@Autowired private CategoryService categoryService;

@GetMapping("/brands")
```

```
public String listFirstPage() {
return defaultRedirectURL;
}
@GetMapping("/brands/page/{pageNum}")
public String listByPage(
@PagingAndSortingParam(listName = "listBrands", moduleURL =
"/brands") PagingAndSortingHelper helper,
@PathVariable(name = "pageNum") int pageNum
) {
brandService.listByPage(pageNum, helper);
return "brands/brands";
}
@GetMapping("/brands/new")
public String newBrand(Model model) {
List<Category> listCategories =
categoryService.listCategoriesUsedInForm();
model.addAttribute("listCategories", listCategories);
model.addAttribute("brand", new Brand());
model.addAttribute("pageTitle", "Create New Brand")
return "brands/brand_form";
}
@PostMapping("/brands/save")
public String saveBrand(Brand brand, @RequestParam("fileImage") MultipartFile
multipartFile,
RedirectAttributes ra) throws IOException {
if (!multipartFile.isEmpty()) {
String fileName =
StringUtils.cleanPath(multipartFile.getOriginalFilename());
```

```
brand.setLogo(fileName);
Brand savedBrand = brandService.save(brand);
String uploadDir = "brand-logos/" + savedBrand.getId();
AmazonS3Util.removeFolder(uploadDir);
AmazonS3Util.uploadFile(uploadDir, fileName,
multipartFile.getInputStream());
} else {
brandService.save(brand);
}
ra.addFlashAttribute("message", "The brand has been saved successfully.");
return defaultRedirectURL;
}
@GetMapping("/brands/edit/{id}")
public String editBrand(@PathVariable(name = "id") Integer id, Model model,
RedirectAttributes ra) {
try {
Brand brand = brandService.get(id);
List<Category> listCategories =
categoryService.listCategoriesUsedInForm();
model.addAttribute("brand", brand);
model.addAttribute("listCategories", listCategories);
model.addAttribute("pageTitle", "Edit Brand (ID: " + id + ")");
return "brands/brand_form";
} catch (BrandNotFoundException ex) {
ra.addFlashAttribute("message", ex.getMessage());
return defaultRedirectURL;
}
}
```

```
@GetMapping("/brands/delete/{id}")
public String deleteBrand(@PathVariable(name = "id") Integer id,
Model model,
RedirectAttributes redirectAttributes) {
try {
brandService.delete(id);
String brandDir = "brand-logos/" + id;
AmazonS3Util.removeFolder(brandDir);
redirectAttributes.addFlashAttribute("message",
"The brand ID " + id + " has been deleted
successfully");
} catch (BrandNotFoundException ex) {
redirectAttributes.addFlashAttribute("message", ex.getMessage());
}
return defaultRedirectURL;
}
}
```

Customer.controller

```
package com.shopme.admin.customer;
import java.util.List;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.servlet.mvc.support.RedirectAttributes;
import com.shopme.admin.paging.PagingAndSortingHelper;
```

```
import com.shopme.admin.paging.PagingAndSortingParam;
import com.shopme.common.entity.Country;
import com.shopme.common.entity.Customer;
import com.shopme.common.exception.CustomerNotFoundException;
@Controller
public class CustomerController {
private String defaultRedirectURL =
"redirect:/customers/page/1?sortField=firstName&sortDir=asc";
@Autowired private CustomerService service;
@GetMapping("/customers")
public String listFirstPage(Model model) {
return defaultRedirectURL;
}
@GetMapping("/customers/page/{pageNum}")
public String listByPage(
@PagingAndSortingParam(listName = "listCustomers",
moduleURL = "/customers") PagingAndSortingHelper helper,
@PathVariable(name = "pageNum") int pageNum) {
service.listByPage(pageNum, helper);
return "customers/customers";
}
@GetMapping("/customers/{id}/enabled/{status}")
public String updateCustomerEnabledStatus(@PathVariable("id") Integer id,
@PathVariable("status") boolean enabled, RedirectAttributes
redirectAttributes) {
service.updateCustomerEnabledStatus(id, enabled);
String status = enabled ? "enabled" : "disabled";
String message = "The Customer ID " + id + " has been " + status;
```

```
redirectAttributes.addFlashAttribute("message", message);
return defaultRedirectURL;
}
@GetMapping("/customers/detail/{id}")
public String viewCustomer(@PathVariable("id") Integer id, Model model,
RedirectAttributes ra) {
try {
Customer customer = service.get(id);
model.addAttribute("customer", customer);
return "customers/customer_detail_modal";
} catch (CustomerNotFoundException ex) {
ra.addFlashAttribute("message", ex.getMessage());
return defaultRedirectURL;
}
}
@GetMapping("/customers/edit/{id}")
public String editCustomer(@PathVariable("id") Integer id, Model model,
RedirectAttributes ra) {
try {
Customer customer = service.get(id);
List<Country> countries = service.listAllCountries();
model.addAttribute("listCountries", countries);
model.addAttribute("customer", customer);
model.addAttribute("pageTitle", String.format("Edit Customer (ID:
%d)", id));
return "customers/customer_form";
} catch (CustomerNotFoundException ex) {
ra.addFlashAttribute("message", ex.getMessage());
```

```
return defaultRedirectURL;
}
}
@PostMapping("/customers/save")
public String saveCustomer(Customer customer, RedirectAttributes ra) {
service.save(customer);
ra.addFlashAttribute("message", "The customer ID " + customer.getId() + "
has been updated successfully.");
return defaultRedirectURL;
}
@GetMapping("/customers/delete/{id}")
public String deleteCustomer(@PathVariable Integer id, RedirectAttributes ra) {
try {
service.delete(id);
ra.addFlashAttribute("message", "The customer ID " + id + " has
been deleted successfully.");
} catch (CustomerNotFoundException ex) {
ra.addFlashAttribute("message", ex.getMessage());
}
return defaultRedirectURL;
}
}
```